



ThinkHub
THINK • LEARN • GROW

The Complete Roadmap to Artificial Intelligence in 2026

Where Beginners Should Start

Presented by

Daksh Khandelwal

Founder, ThinkHub



The 2026 Imperative: Beyond Software Engineering

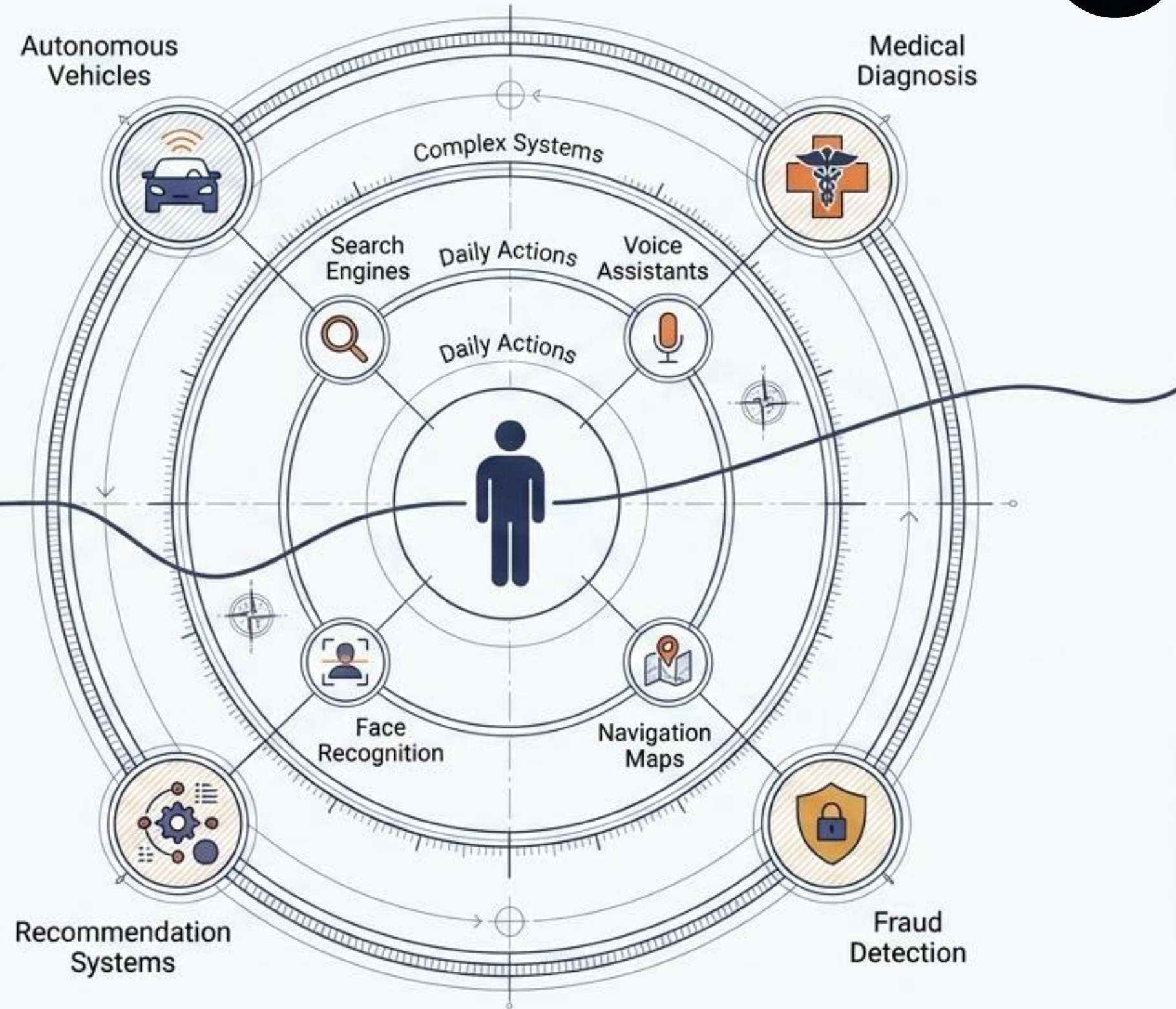
AI is fundamentally changing how the world solves problems. Companies are investing billions, and the demand for AI fluency continues to rise across all roles.



Key Insight: Doctors, teachers, business analysts, designers, and creators are using AI daily. AI literacy is now a baseline professional skill, regardless of your industry.

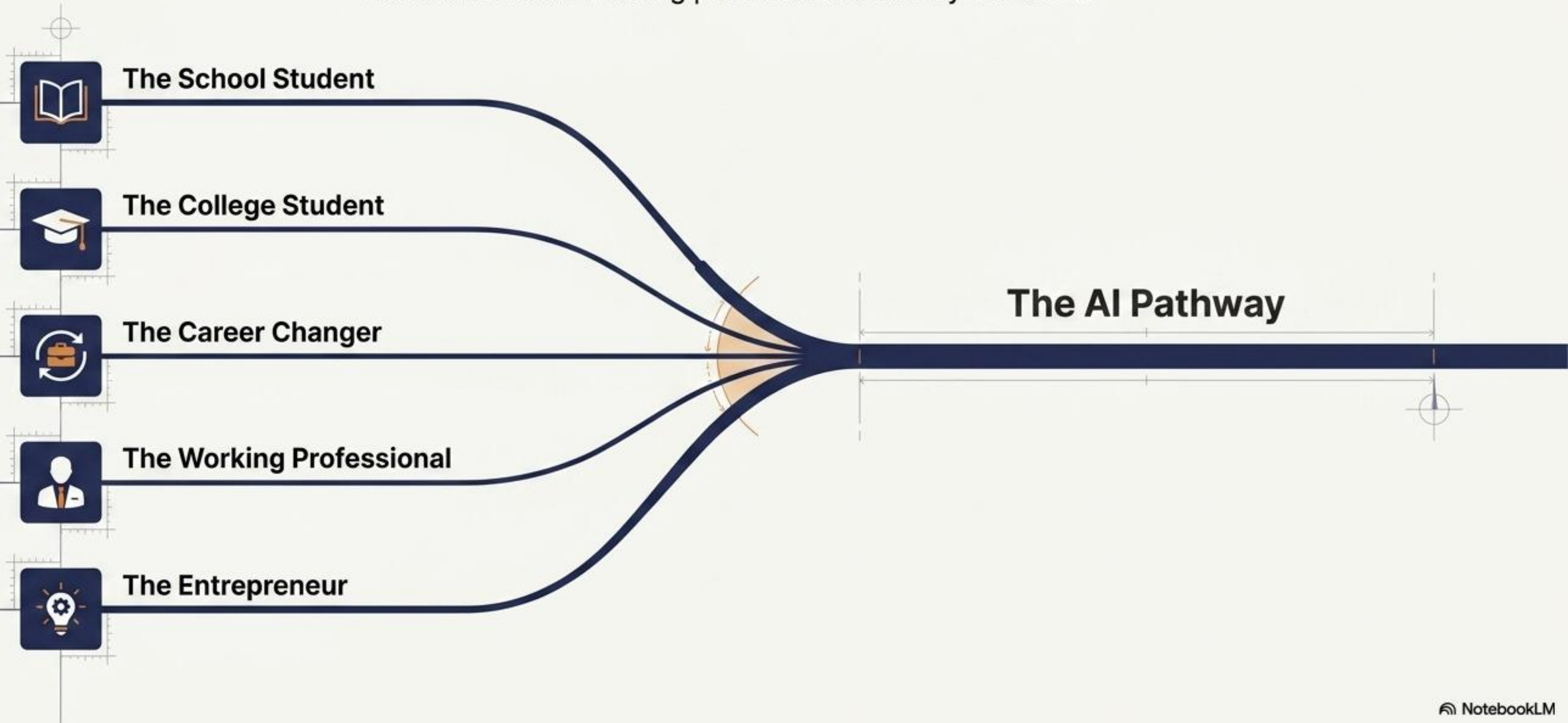
You Are Already Surrounded

Most people use AI every single day without realizing it. From spam filters to personalized shopping, you are already an end-user. Now, it's time to become a creator.



All Paths Lead Forward

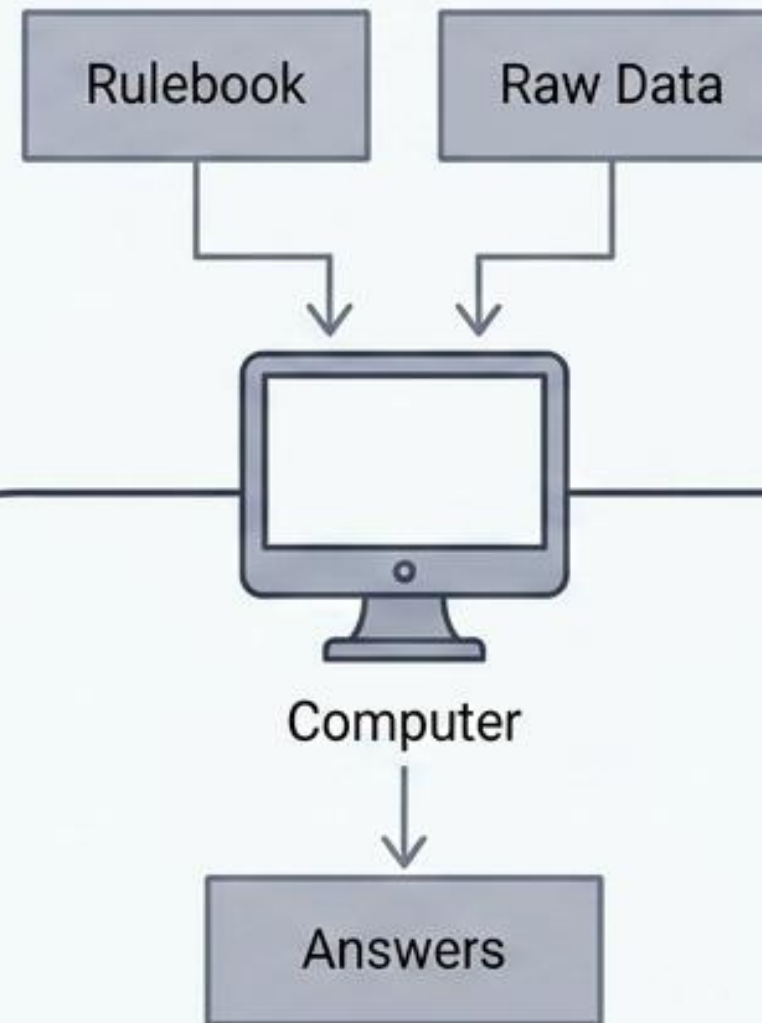
Whether you are starting from absolute zero or pivoting from an existing career, the foundational learning path remains exactly the same.



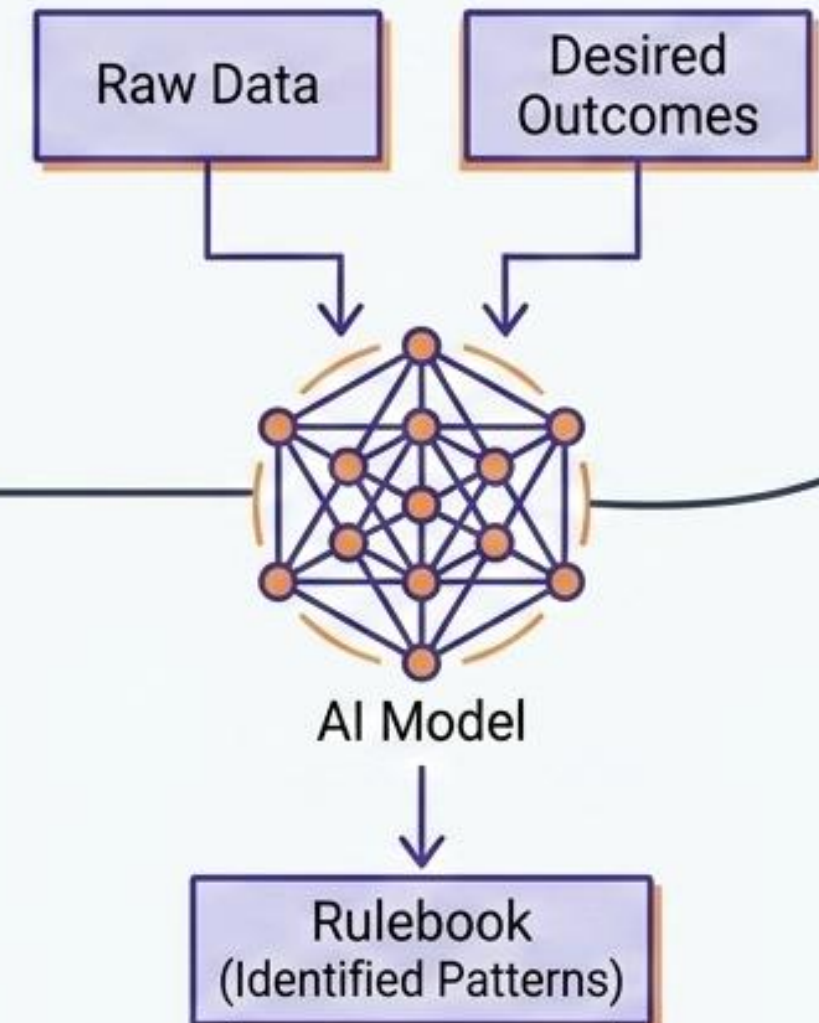
The Paradigm Shift: What is AI, Really?

- AI enables machines to perform tasks requiring human-like intelligence.
- It doesn't 'think'—it learns from data, identifies patterns, and makes predictions to support decision-making.

Traditional Programming

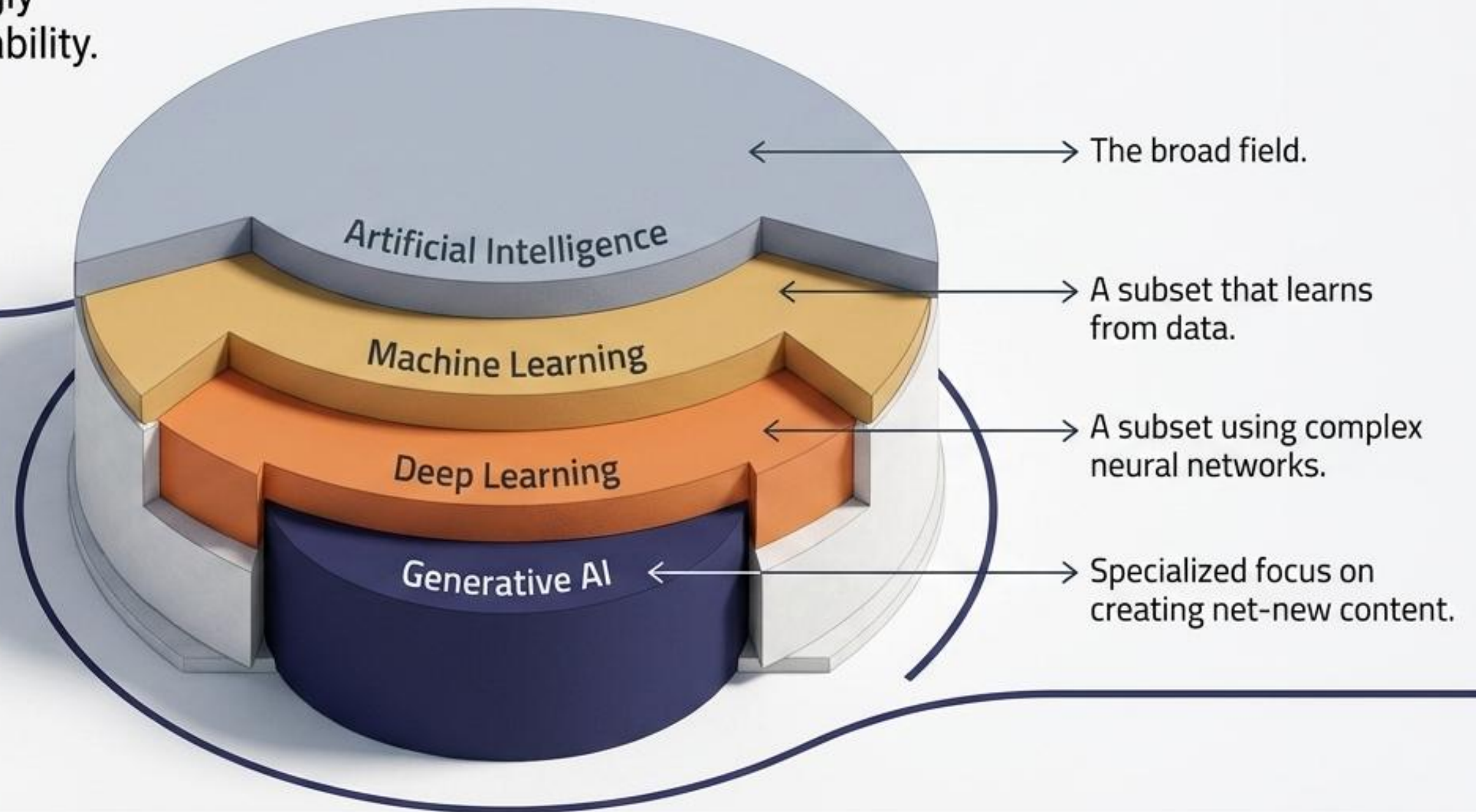


Artificial Intelligence



The AI Ecosystem Hierarchy

The terms are not interchangeable.
They represent increasingly
specialized layers of capability.

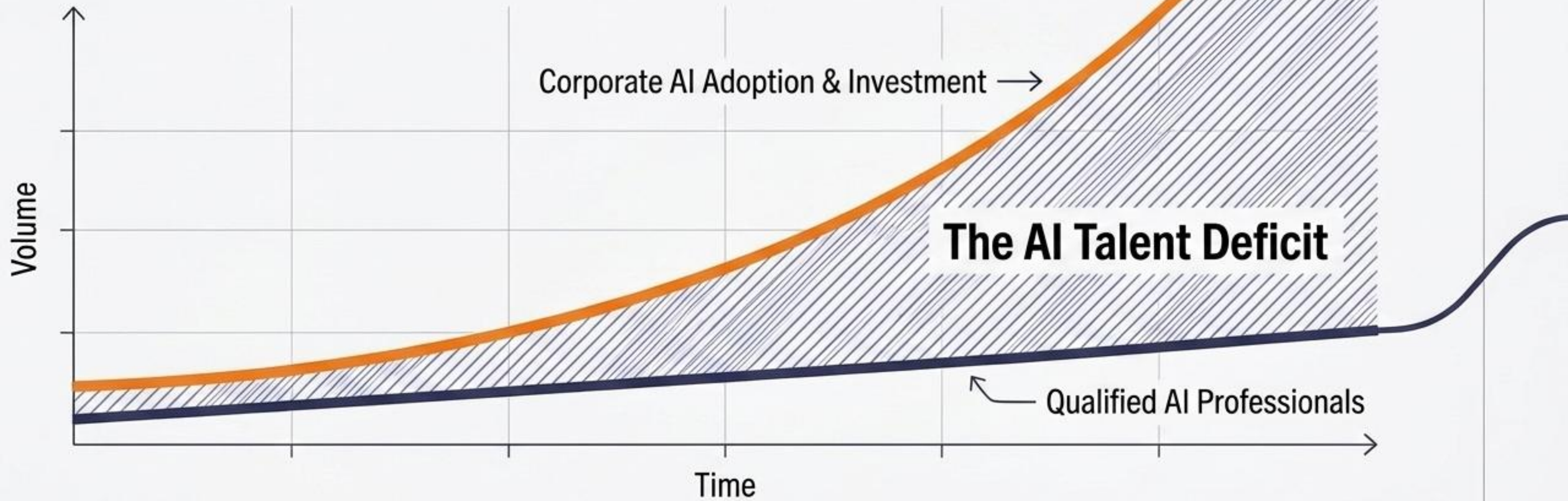


How the Ecosystem Actually Works

		The Input	The Process	The Output
	Artificial Intelligence	Pre-defined Rules	Logic & Search	Automates Tasks
	Machine Learning	Structured Data	Statistical Learning	Predicts Outcomes
	Deep Learning	Unstructured Data (Audio/Video)	Multi-layer Neural Networks	Understands Context
	Generative AI	Human Prompts	Foundation Models	Creates Net-New Content

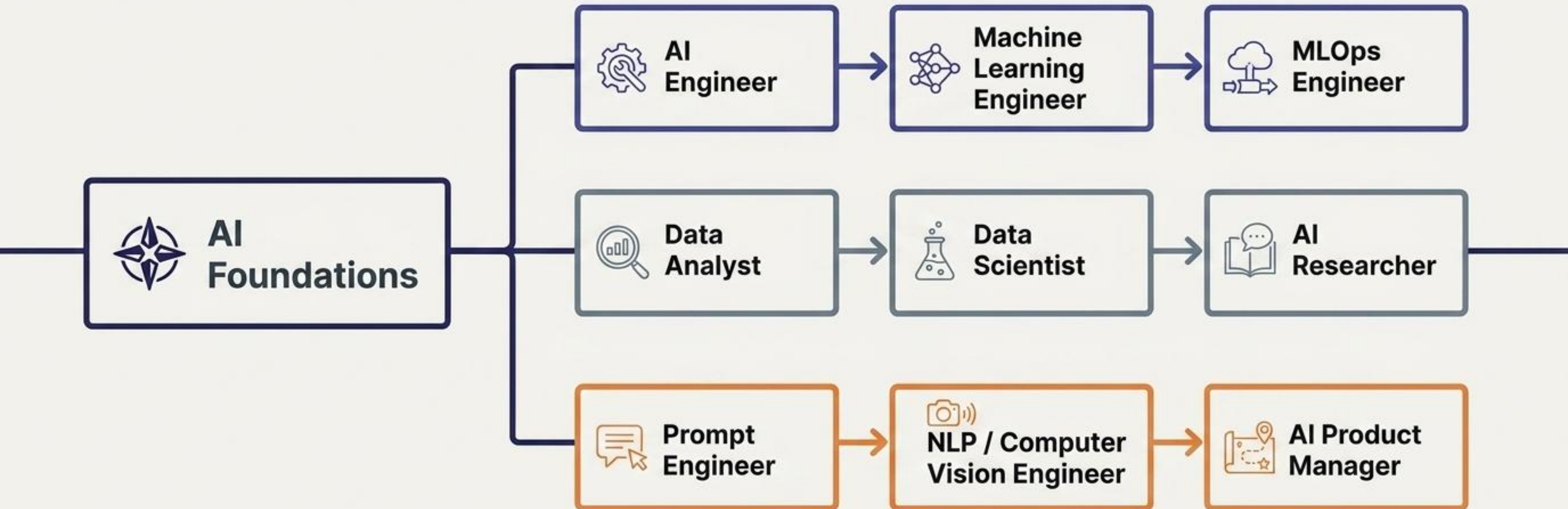
The True Opportunity: The Talent Deficit

The gap between rapid business adoption and the supply of AI-literate professionals is the exact space your future career will occupy.



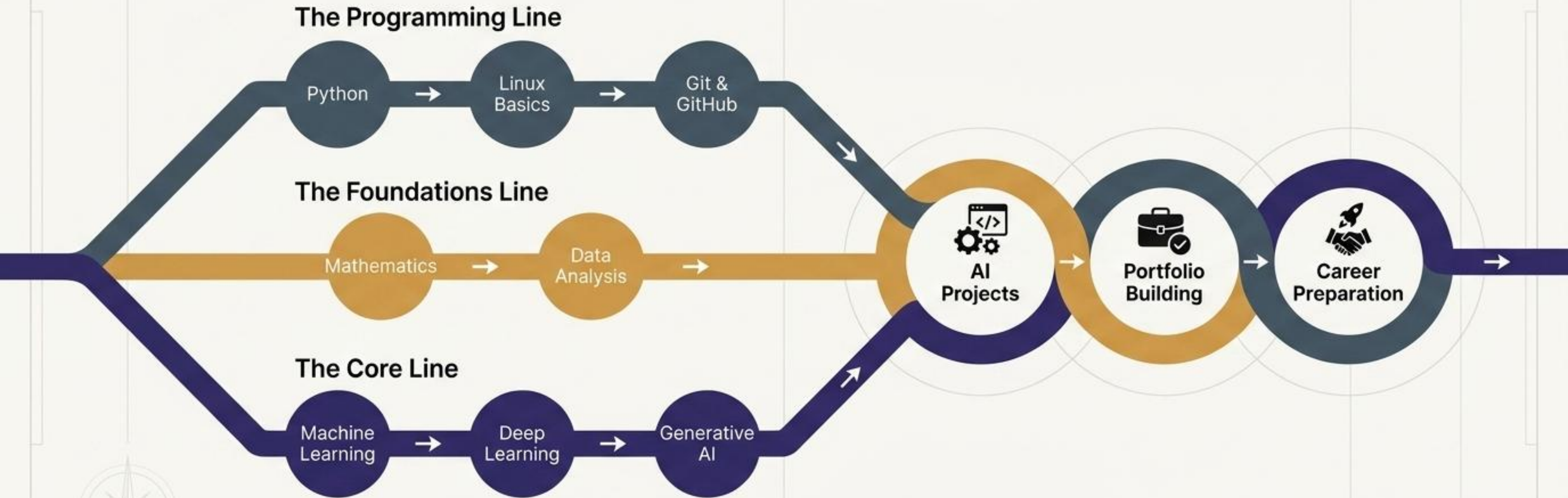
Mapping Your Future Role

AI is not a single job title. Your foundations will allow you to branch into building systems, analyzing data, or managing AI-powered products.



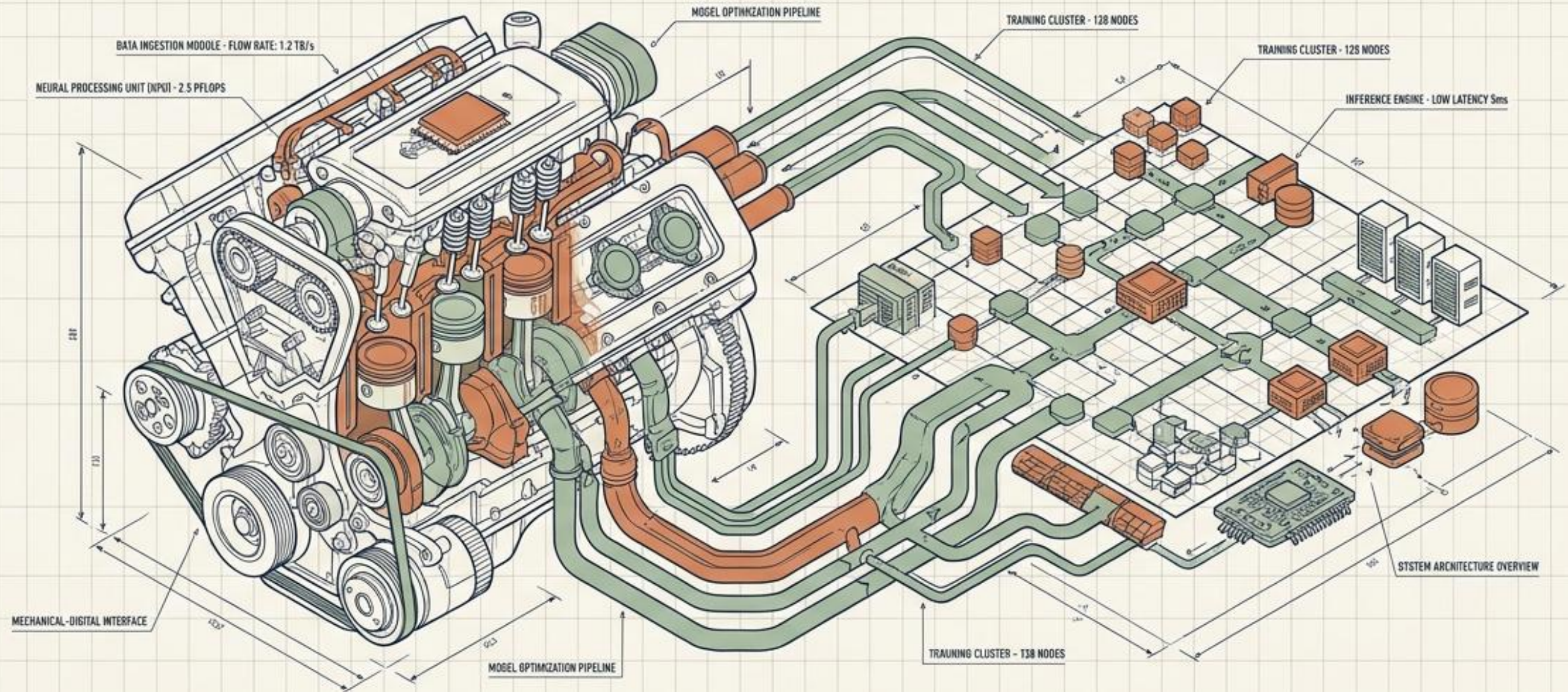
The Complete 2026 AI Learning Roadmap

The exact sequence of skills you need to bridge the talent deficit. (Take a screenshot).



THE AI ENGINEER BLUEPRINT

Core Foundations



The Foundational Journey

The Vehicle
Python & Essential
Libraries



The Environment
Git & Linux



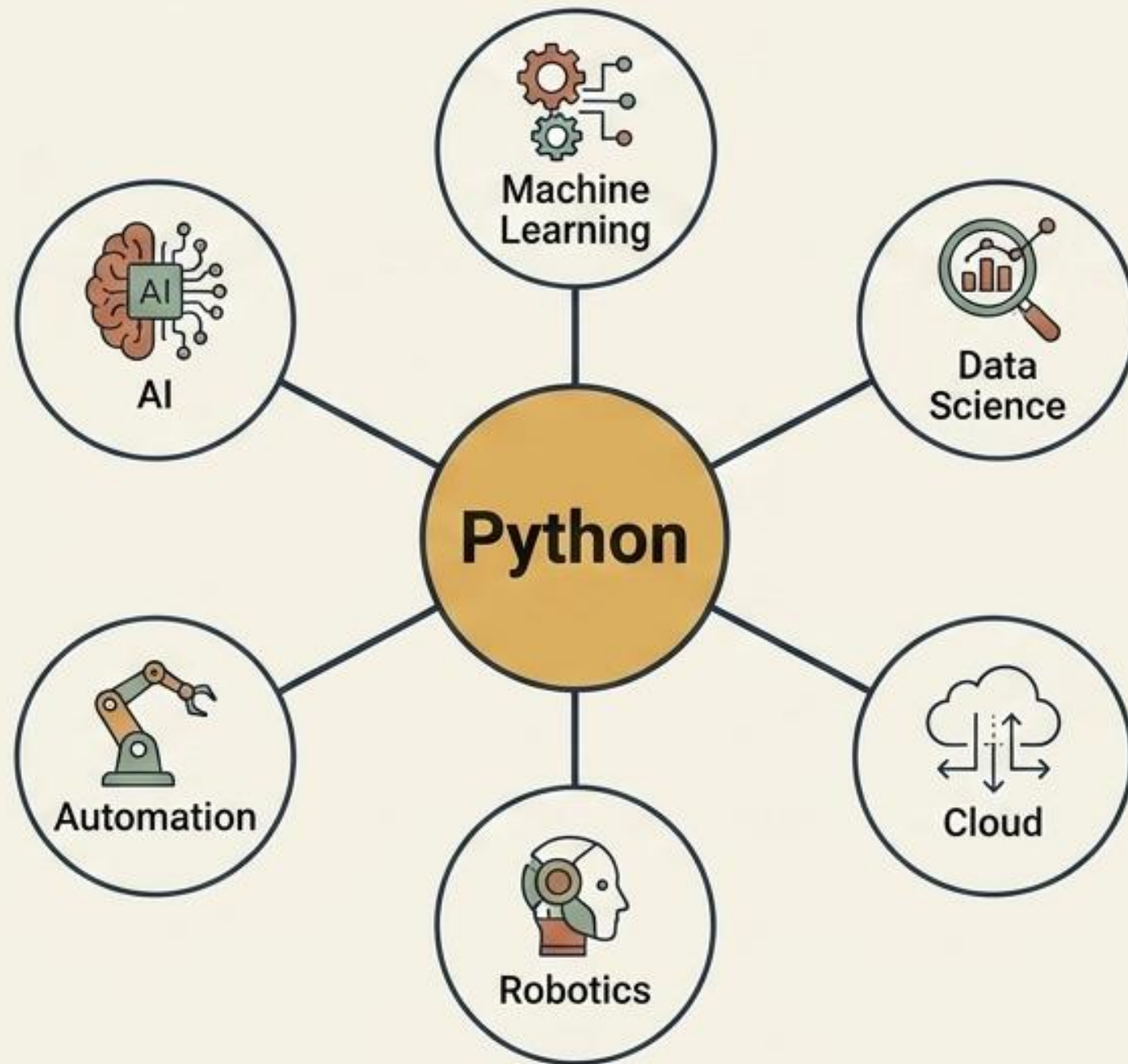
The Engine
Mathematics



The Fuel
Data Analysis &
Visualization



Why Python Drives Artificial Intelligence



Why not C++ or Java?

In 2026, Python has the richest ecosystem of AI libraries, the simplest syntax, and the largest learning community.



Language of AI

(Most libraries are built for it)



Beginner-Friendly

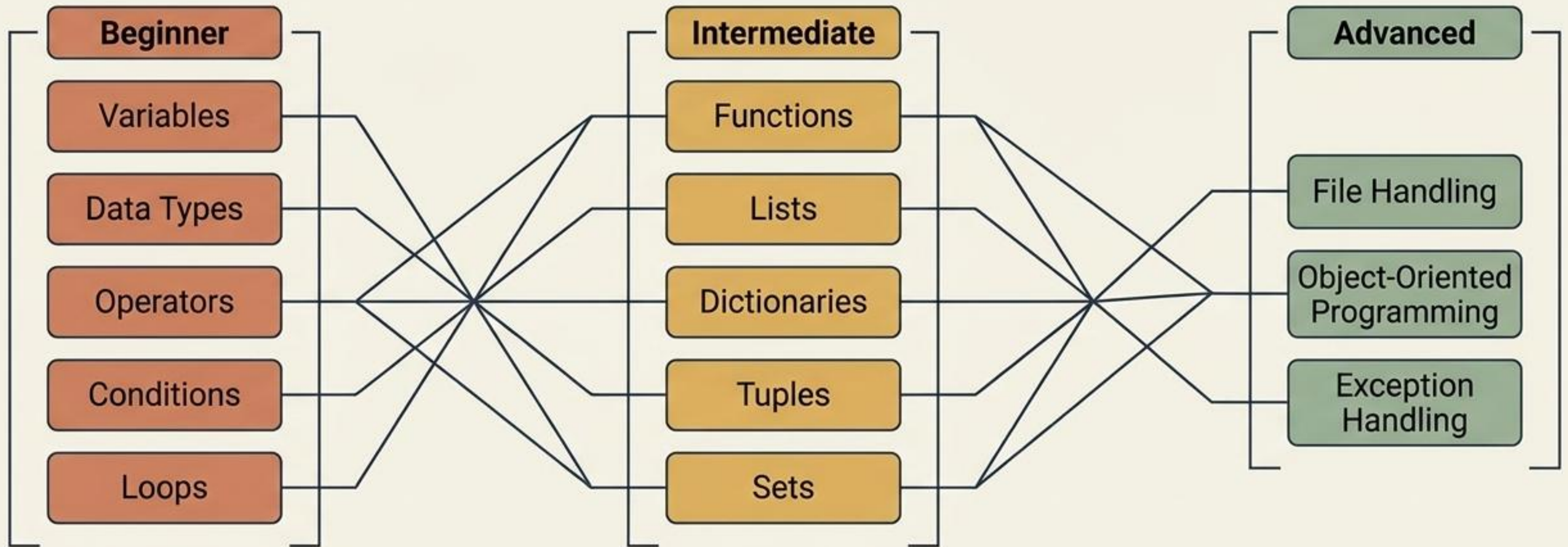
(Easy syntax to learn fast)



Massive Community

(Unmatched support and resources)

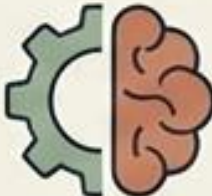
The Python Skill Tree



You don't need to master every feature. Focus on concepts used in AI projects.

Recommended Resources: Official Python Documentation | Harvard CS50 (Python) | freeCodeCamp Python Course.

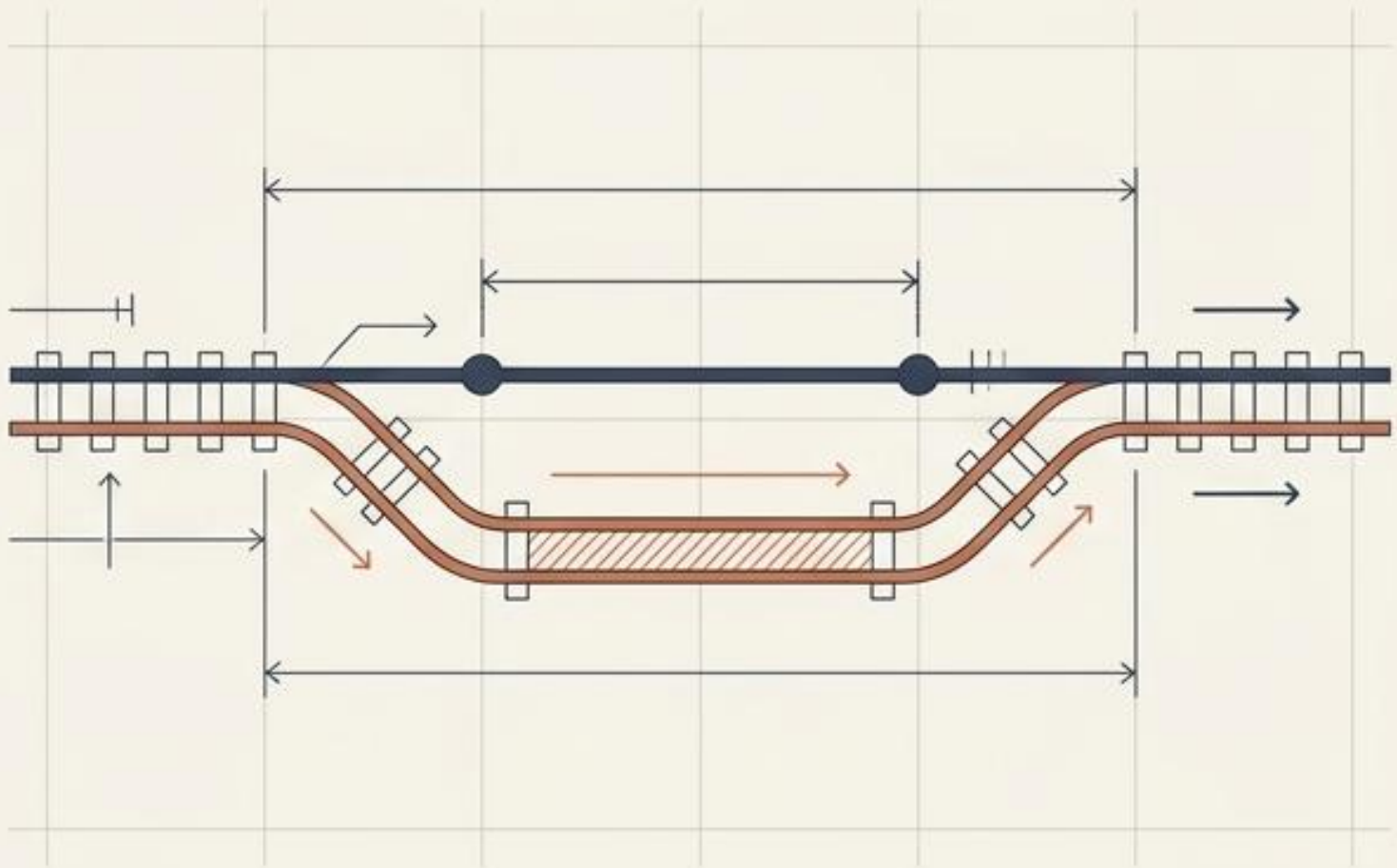
The Essential AI Toolbox

Data Manipulation		Data Visualization			
 NumPy	 Pandas	 Matplotlib	 Plotly		
Classic Machine Learning		Deep Learning & Vision			
 Scikit-learn		 TensorFlow	 PyTorch	 OpenCV	 Hugging Face

Think of Python as the toolbox. These libraries are the specialized tools for mathematics, analysis, and deep learning.

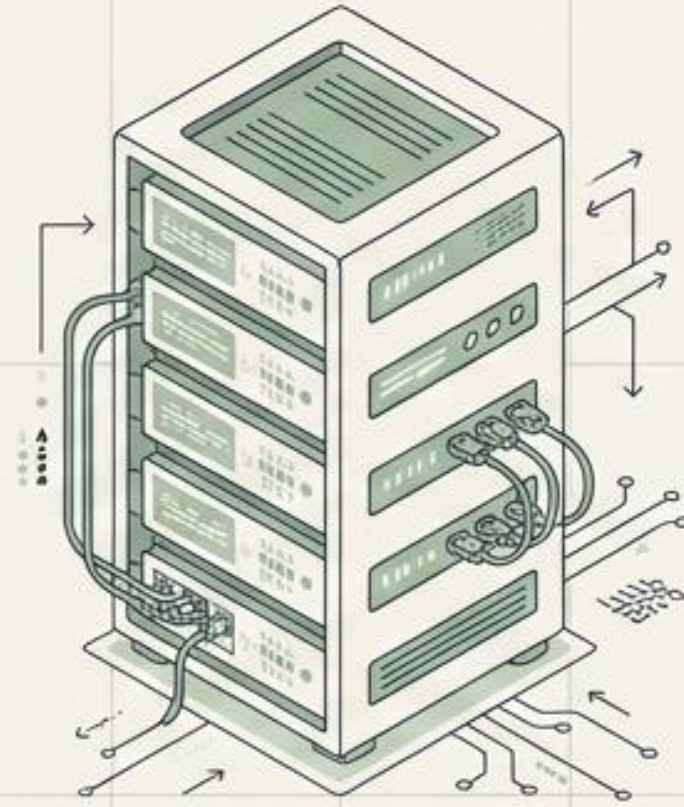
Setting Up The Professional Workspace

Git's Parallel Universes



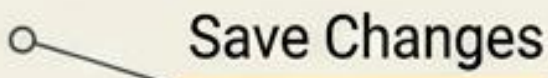
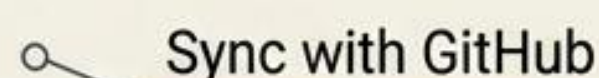
Track changes, collaborate, and build a portfolio
recruiters will actually check.

The Linux Bedrock



Most AI servers, cloud machines, and supercomputers run Linux.
Basic command-line skills make your workflow seamless.

The Command Line Cheatsheet

Linux Workspace	Git Version Control
<code>pwd / ls / cd</code>  <u>Navigation</u>	<code>git init / git clone</code>  <u>Setup</u>
<code>mkdir / rm / cp / mv</code>  <u>File Management</u>	<code>git add / git commit</code>  <u>Save Changes</u>
<code>python / pip</code>  <u>Execution & Packages</u>	<code>git push / git pull</code>  <u>Sync with GitHub</u>



Local Code



Version Control

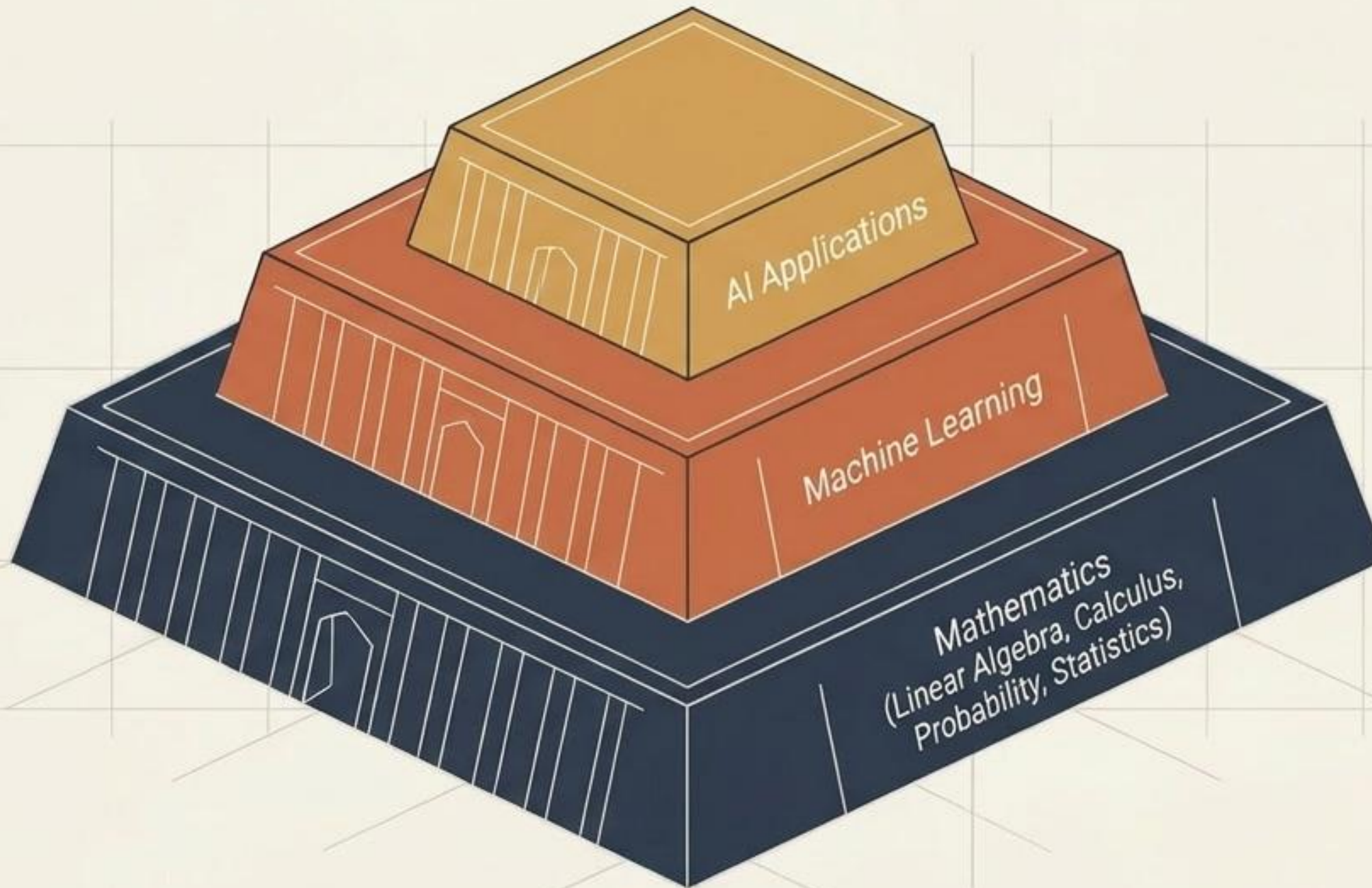


Remote Repository



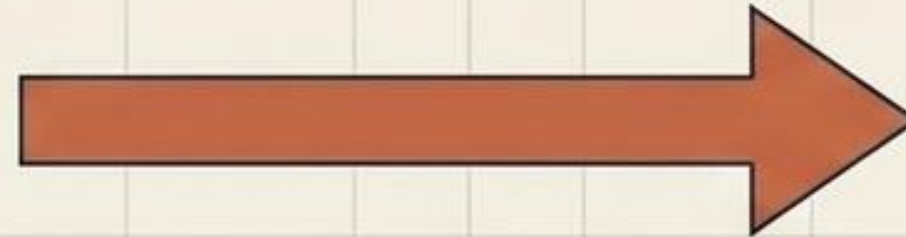
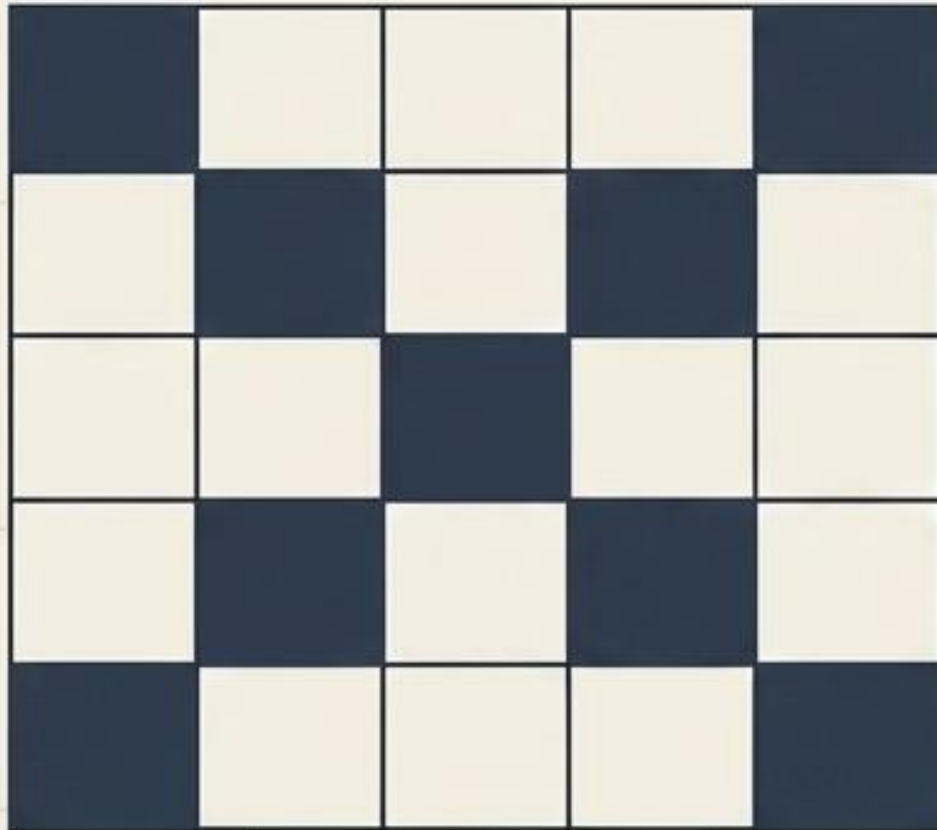
Recruiters

The Invisible Mechanics



Don't let math stop you. Begin with programming and learn mathematical concepts alongside your journey. Focus on intuition first, not memorizing formulas.

Linear Algebra Translated



Computers "see" data as vectors and matrices.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Core Concepts

- Scalars, Vectors, Matrices, Matrix Multiplication, Dot Product, Eigenvalues

Applications

- Neural Networks, Computer Vision, Recommendation Systems

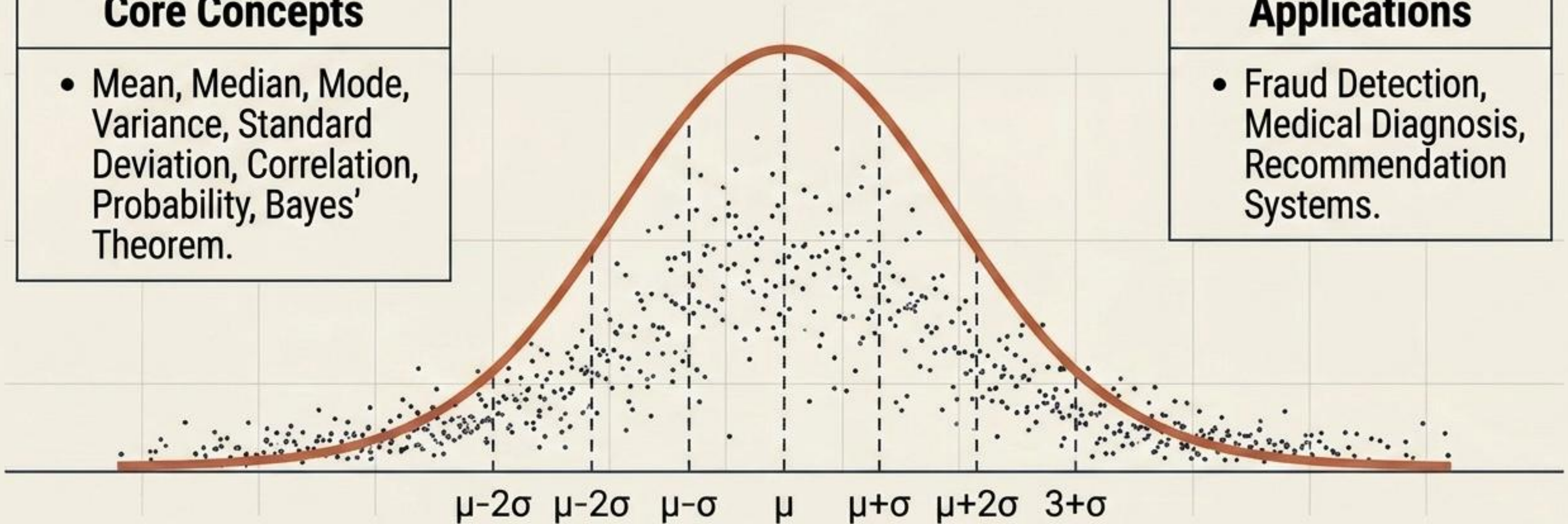
Probability & Statistics

Core Concepts

- Mean, Median, Mode, Variance, Standard Deviation, Correlation, Probability, Bayes' Theorem.

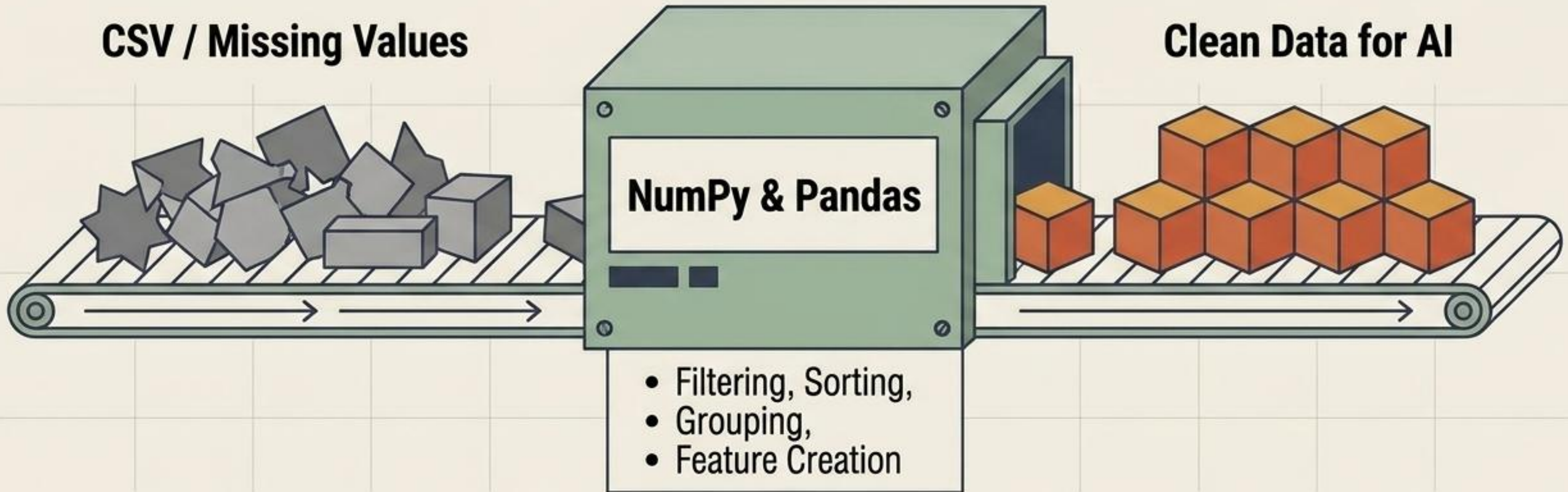
Applications

- Fraud Detection, Medical Diagnosis, Recommendation Systems.



AI doesn't make perfect decisions—it makes probable decisions based on data.

Refining Raw Data

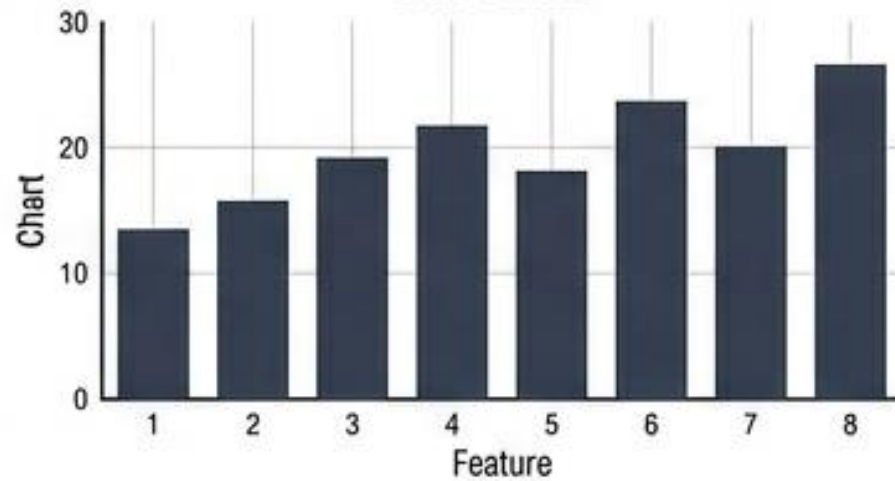


70–80% of an AI engineer's time is spent preparing data. Clean data is essential for model performance.

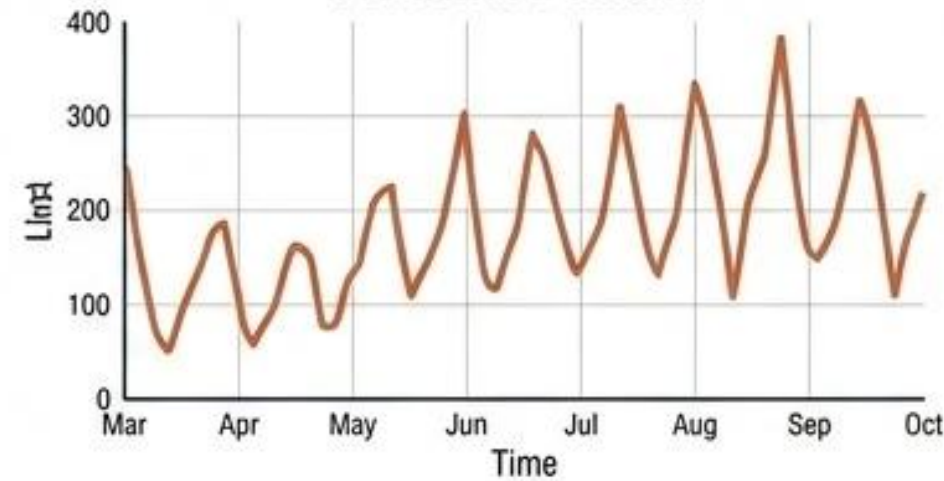
Student Marks Dataset → Remove Missing Values → Calculate Average → Predict Performance.

Communicating Insights

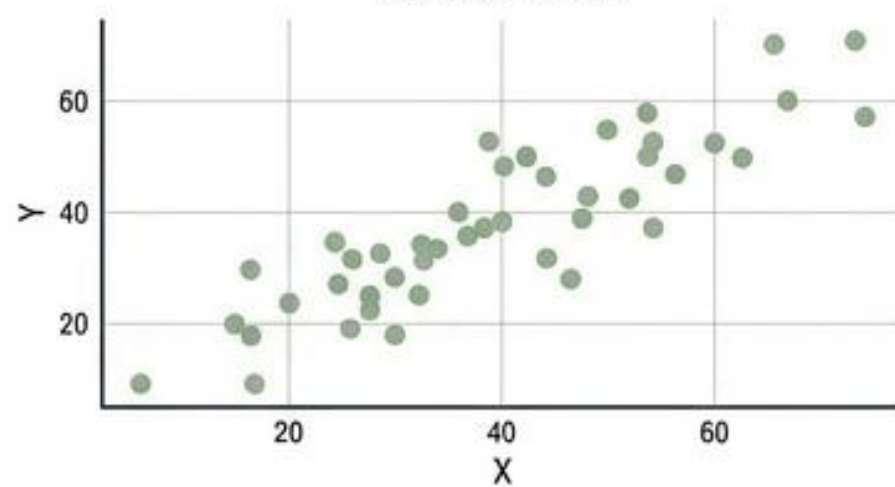
Bar Chart



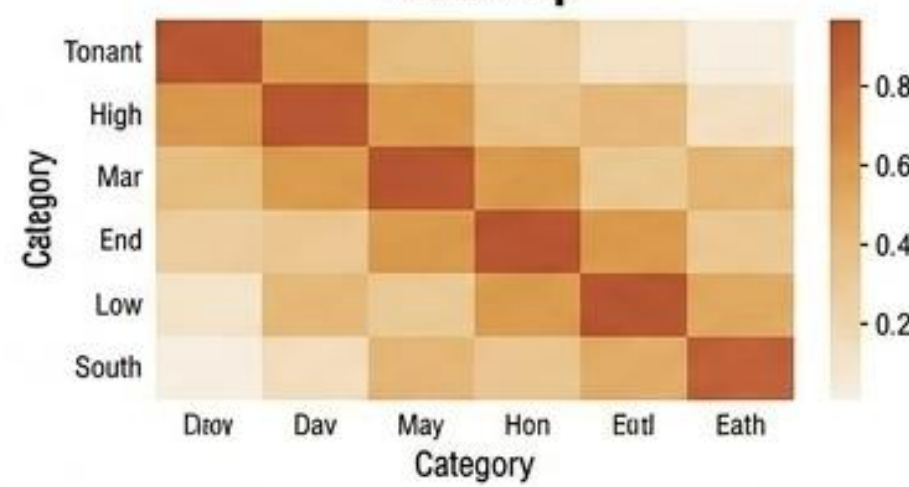
Seasonal Demand



Scatter Plot



Heatmap

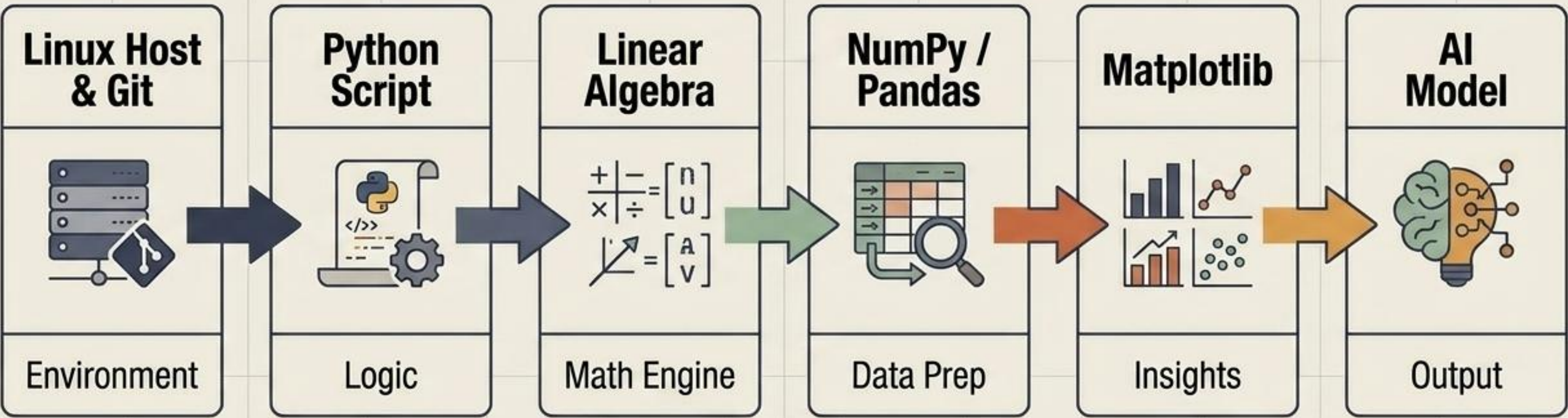


Visualize to understand patterns before training a model.
Good visualization reveals trends invisible in raw data.

Powered by
Matplotlib & Plotly

- Bar Charts,
- Line Charts,
- Scatter Plots,
- Histograms,
- Heatmaps,
- Box Plots.

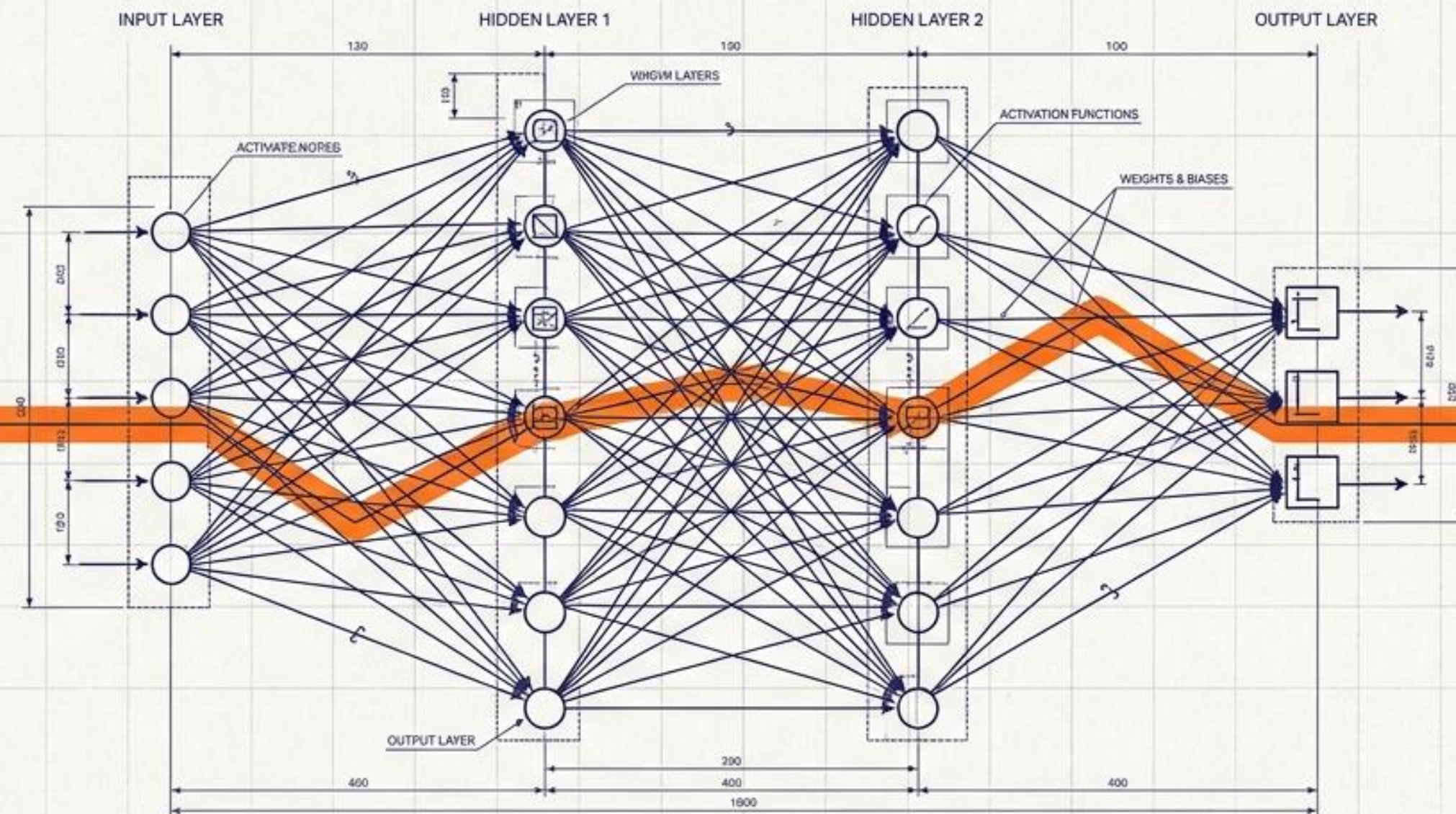
The AI Development Lifecycle



These aren't isolated topics. They are an interconnected ecosystem where every tool plays a specific, necessary role.

THE HEART OF AI

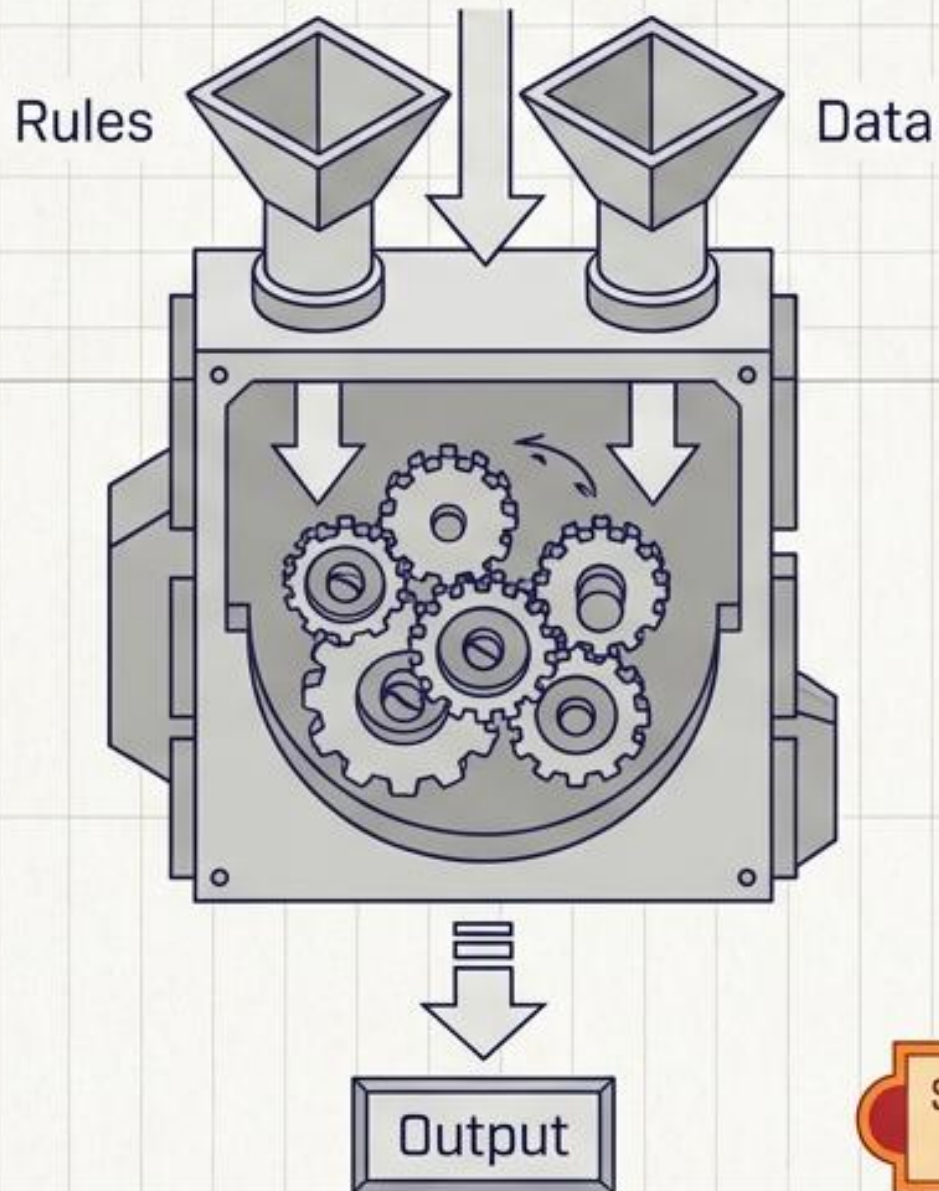
Unlocking Machine Learning and Deep Learning



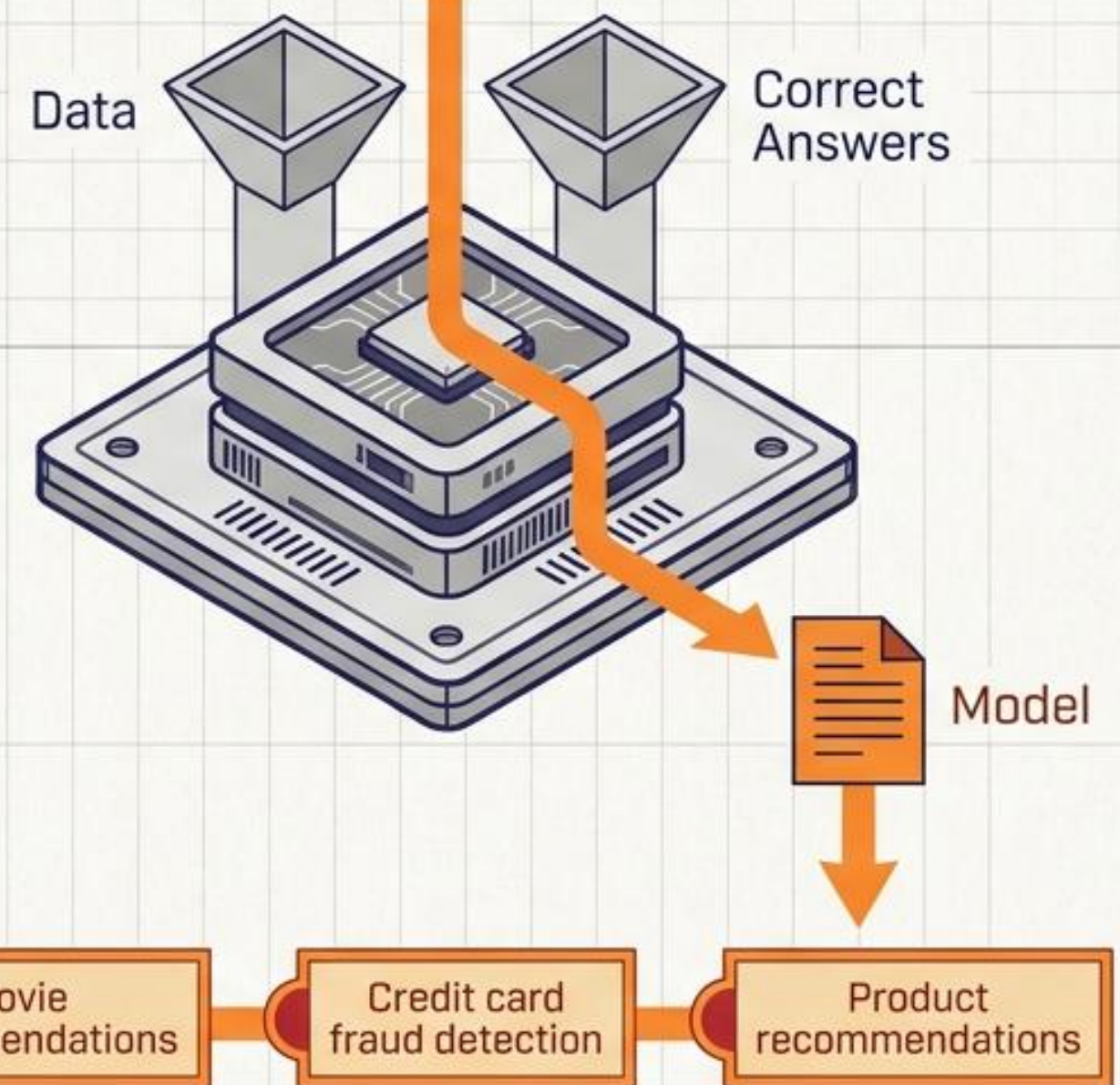
The Paradigm Shift: From Rules to Patterns

Machines learn patterns from data instead of explicit rules.
More data improves model performance.

Traditional Programming

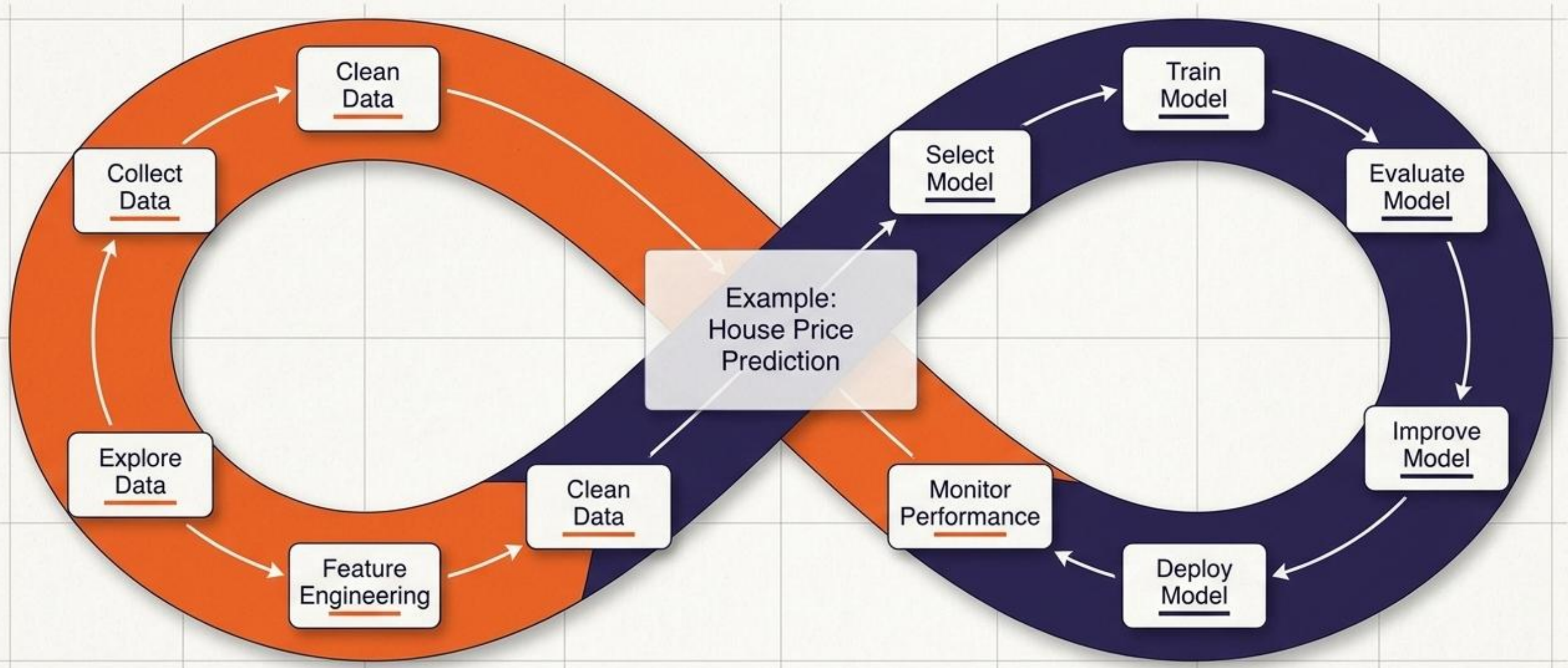


Machine Learning



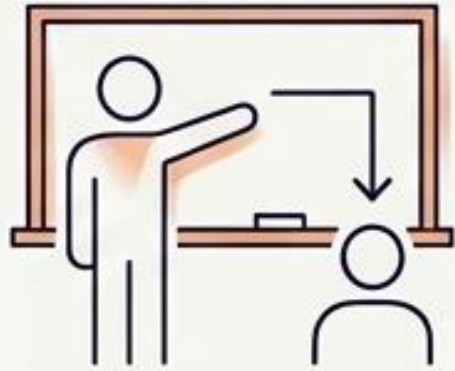
The Machine Learning Lifecycle

Building AI requires more than choosing an algorithm. The majority of real-world effort is spent collecting and understanding data before training begins.



The Machine Learning Trinity

Three unique approaches, chosen based on your data and your objective.



Supervised Learning

House price prediction

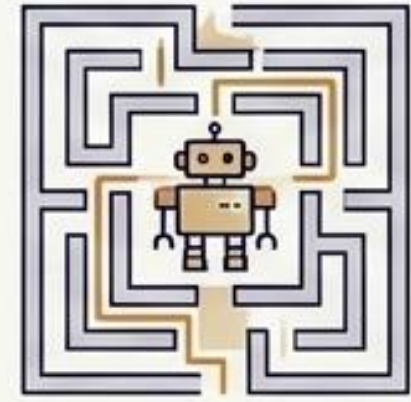
Email spam detection



Unsupervised Learning

Customer segmentation

Market basket analysis



Reinforcement Learning

Chess

Robotics

Self-driving systems

The Editorial Blueprint

Core Algorithms

Linear Regression

Logistic Regression

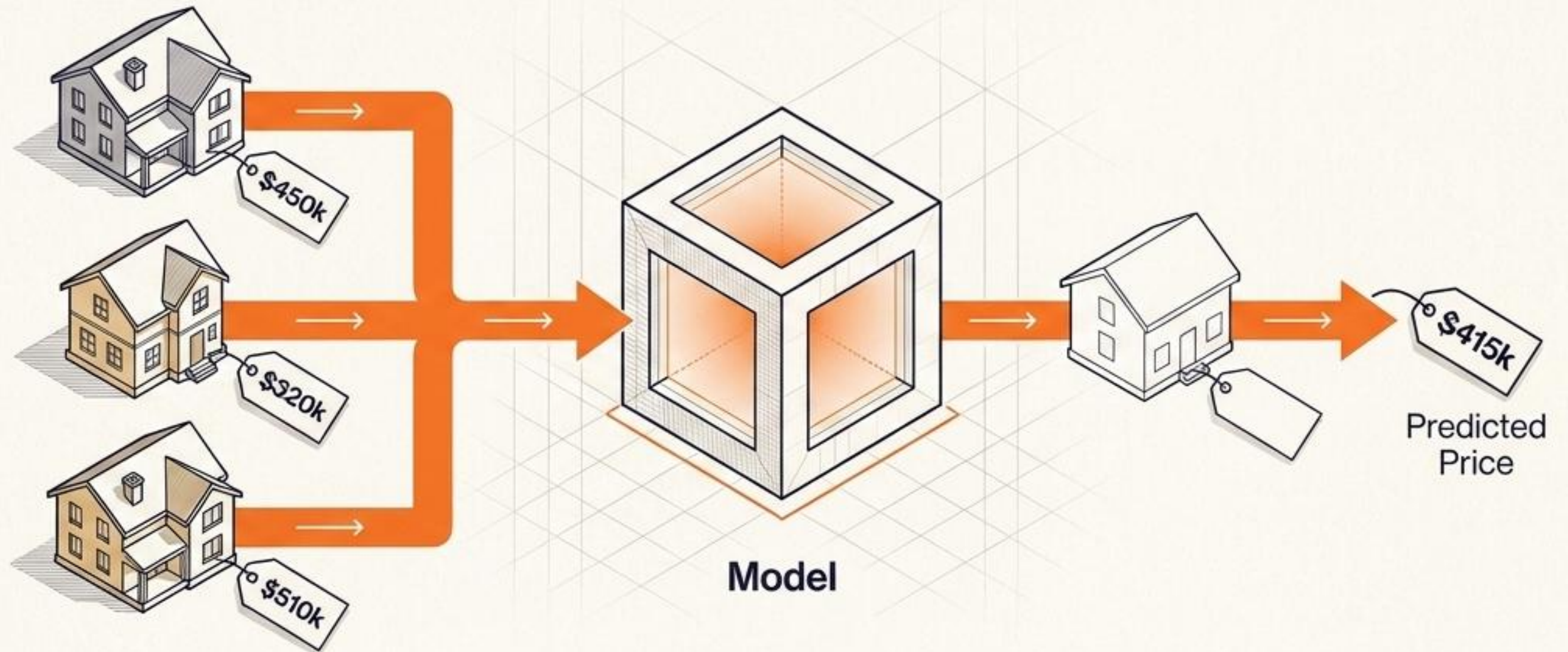
Decision Trees

Random Forest

Support Vector
Machine

Supervised Learning: Learning by Example

Learns from input-output pairs to predict future outcomes using labeled data.



Applications

Medical diagnosis

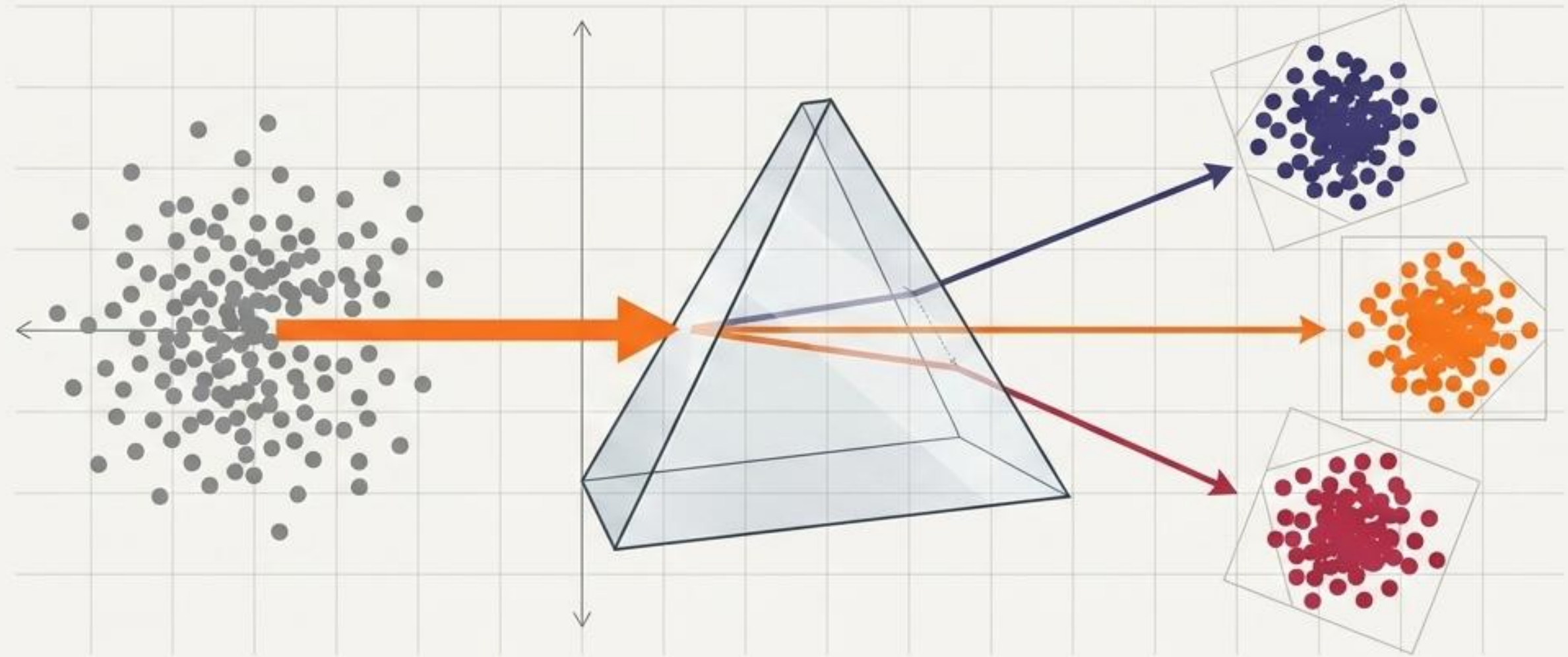
Loan approval

Student performance prediction

Sales forecasting

Unsupervised Learning: Uncovering Hidden Structures

No labeled outputs. The AI automatically finds hidden relationships and groups similar data.



Core Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- PCA (Dimensionality Reduction)

Applications

Customer segmentation

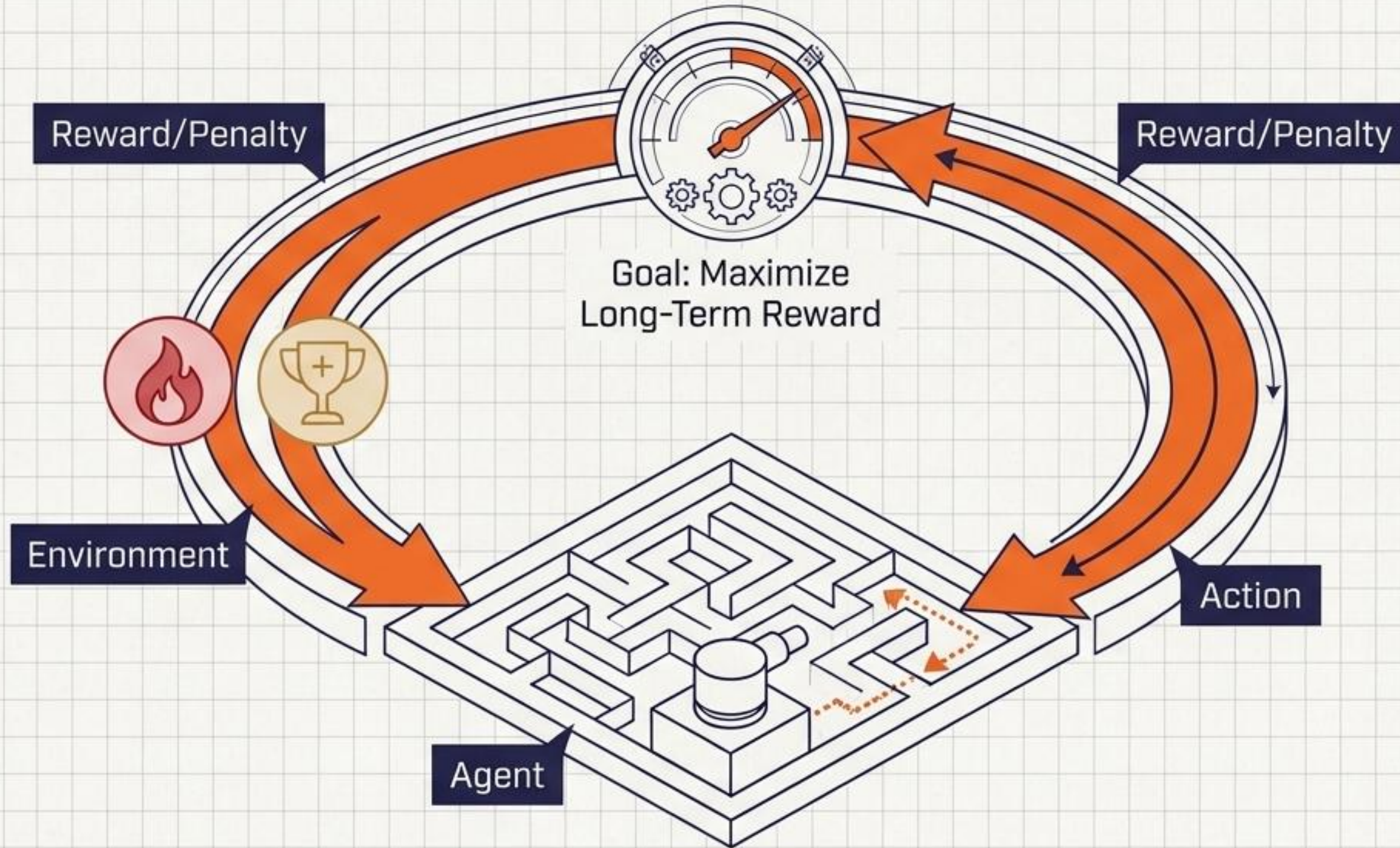
Recommendation systems

Anomaly detection

Topic discovery

Reinforcement Learning: The Reward Loop

Teaching AI through continuous trial and error in dynamic environments.



Robotics



Game-playing
AI



Traffic
optimization



Autonomous
vehicles

The Algorithm Toolbelt

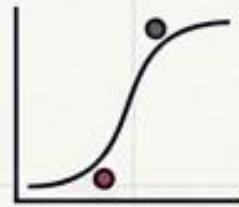
Start with foundational algorithms and expand your toolkit as your projects demand.

Linear Regression



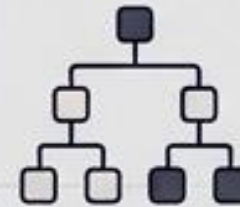
Predicting continuous values

Logistic Regression



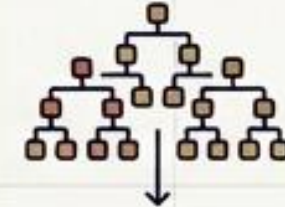
Binary classification

Decision Tree



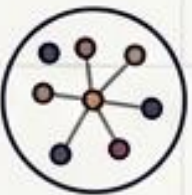
Easy-to-interpret decisions

Random Forest



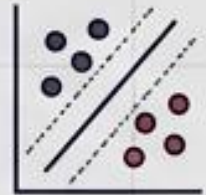
Better accuracy and robustness

K-Nearest Neighbors



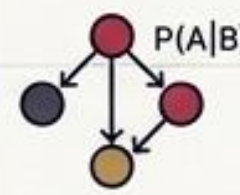
Similarity-based prediction

Support Vector Machine



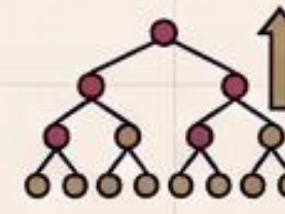
Complex classification

Naïve Bayes



Text classification

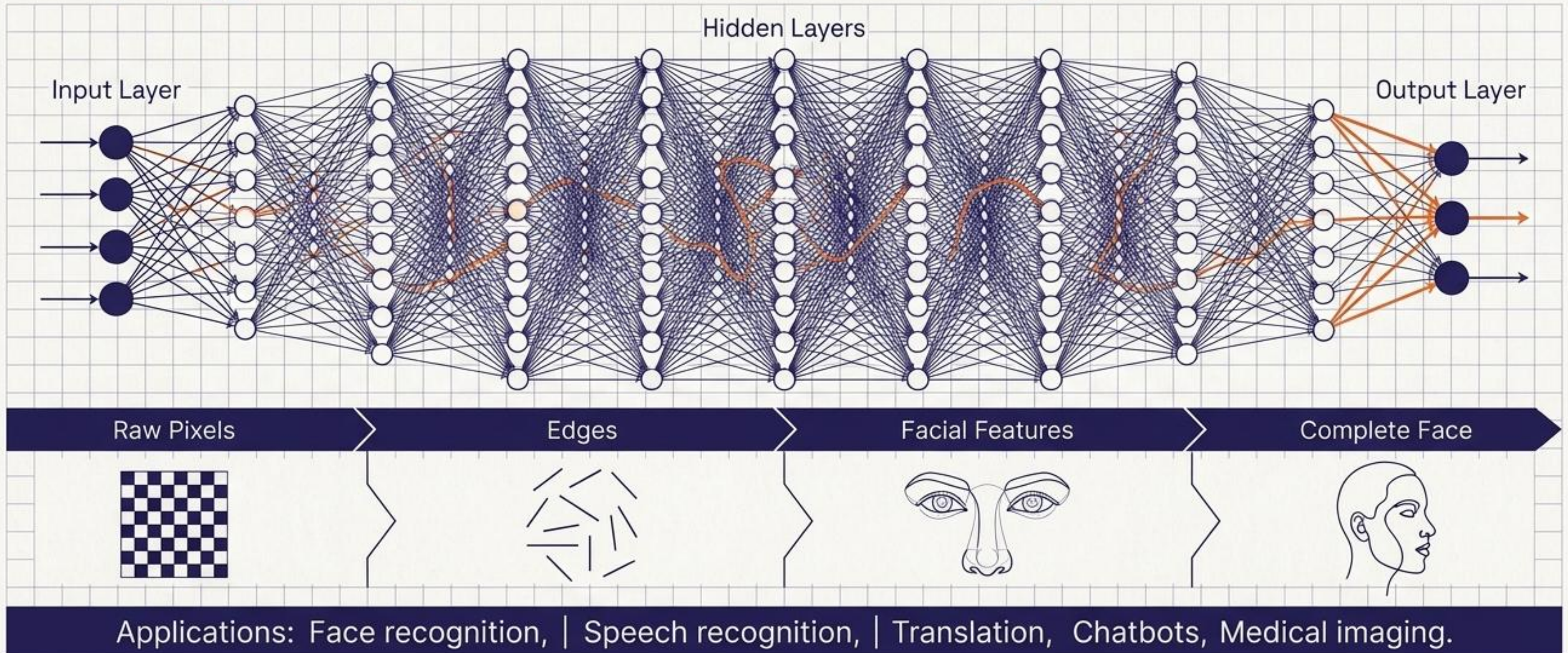
XGBoost



High-performance predictive modeling

The Next Frontier: Enter Deep Learning

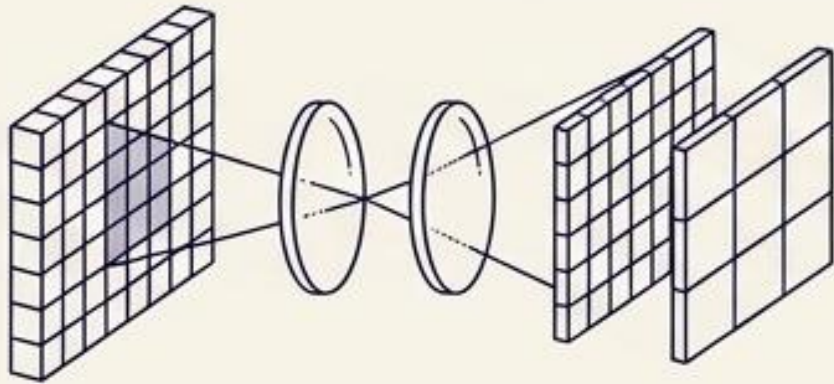
Inspired by the human brain. Deep learning requires massive datasets and computing power to process unstructured data like images, audio, and text.



The Architecture of Deep Learning

Architectures are specialized by data type: vision, sequence, and context.

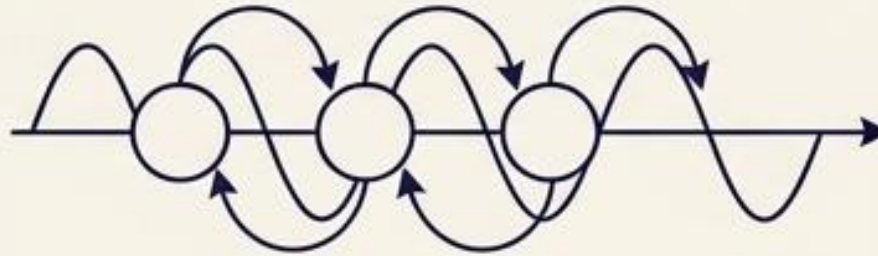
CNNs (Convolutional Neural Networks)



Data Shape: Images

Applications: Image classification, Object detection, Medical imaging

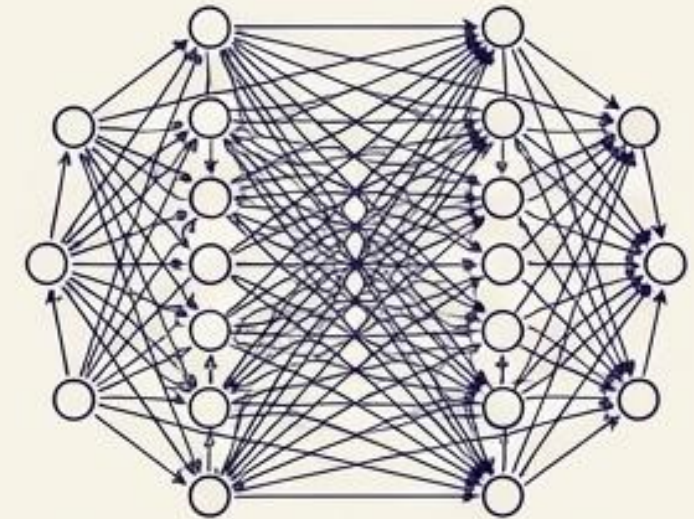
RNNs (Recurrent Neural Networks)



Data Shape: Sequential/Time

Applications: Time series forecasting, Speech recognition

Transformers

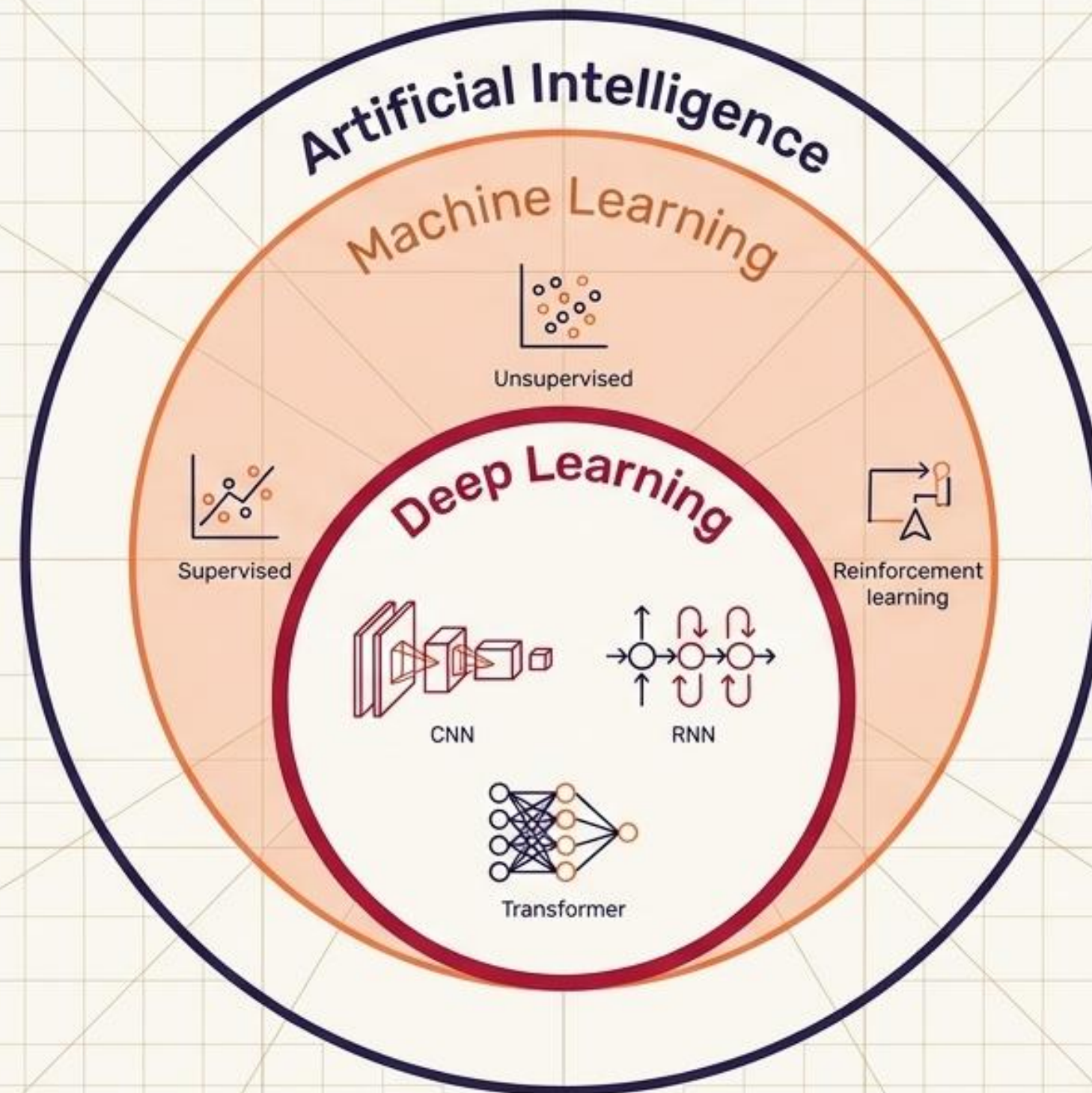


Data Shape: Language/Context

Applications: Language models, Translation, ChatGPT, Claude, Gemini

The AI Universe

How the entire ecosystem connects: Deep Learning is the specialized core powering modern Generative AI.



Taking Action: Your First Blueprint

Real progress happens when you build. Complete the workflow, document your work, and publish it.



1. Choose a dataset

2. Explore & Clean

3. Train & Evaluate

4. Improve & Save

House Price Prediction

Student Performance Prediction

Iris Flower Classification

Movie Recommendation System

The Complete Roadmap to Artificial Intelligence in 2026

ThinkGov Masterclass Series

From foundational theory to a hired portfolio in 90 days.



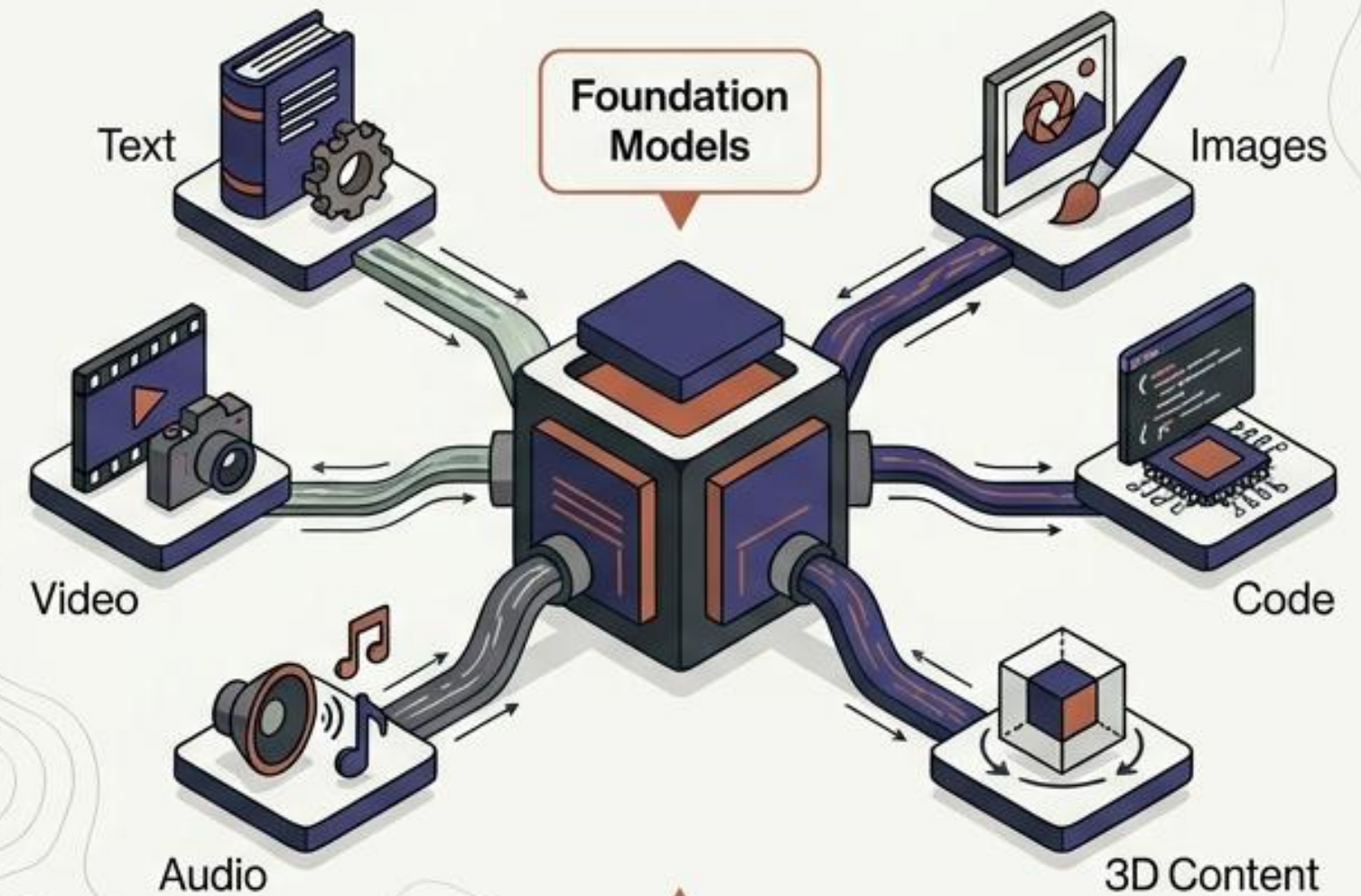
The Generative era shifts artificial intelligence from analysis to creation.

The Past: Analyze & Predict



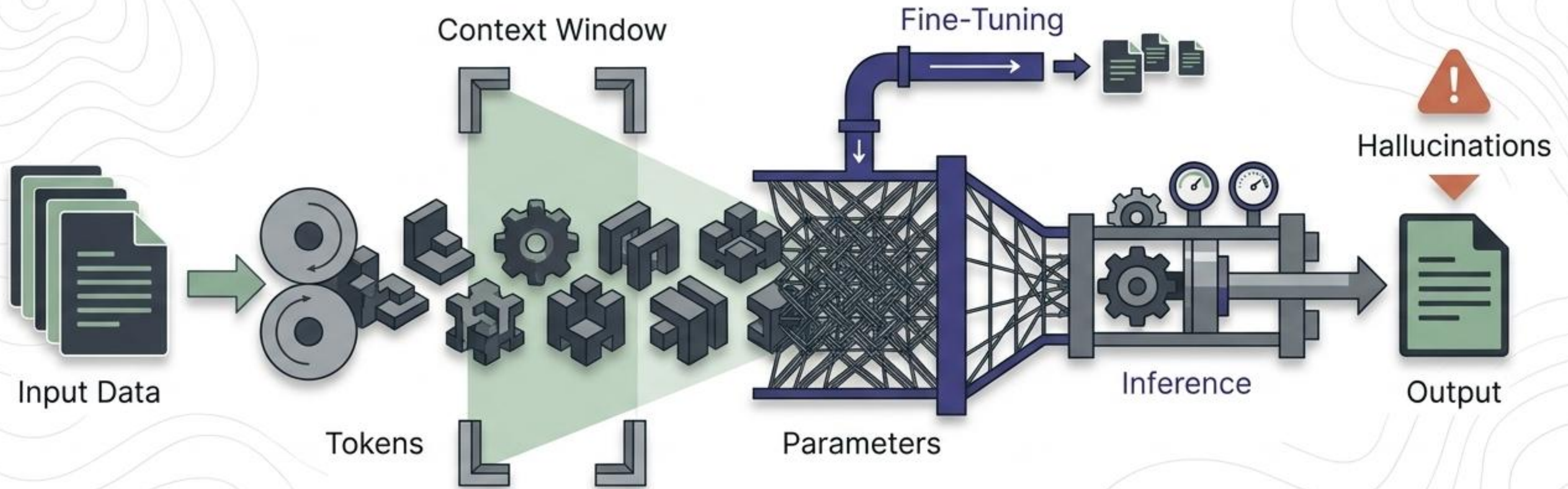
Focuses on understanding existing data to make calculated forecasts and optimize established processes.

The Present: Create & Synthesize



Real-World Engines: ChatGPT, Claude, Gemini, Midjourney, GitHub Copilot, Suno AI.

Large Language Models process text through structured mathematical compression.



Leading Models: GPT, Claude, Gemini, Llama, Mistral, Qwen.

The evolution of AI capabilities demands a shift from prompting to delegating.

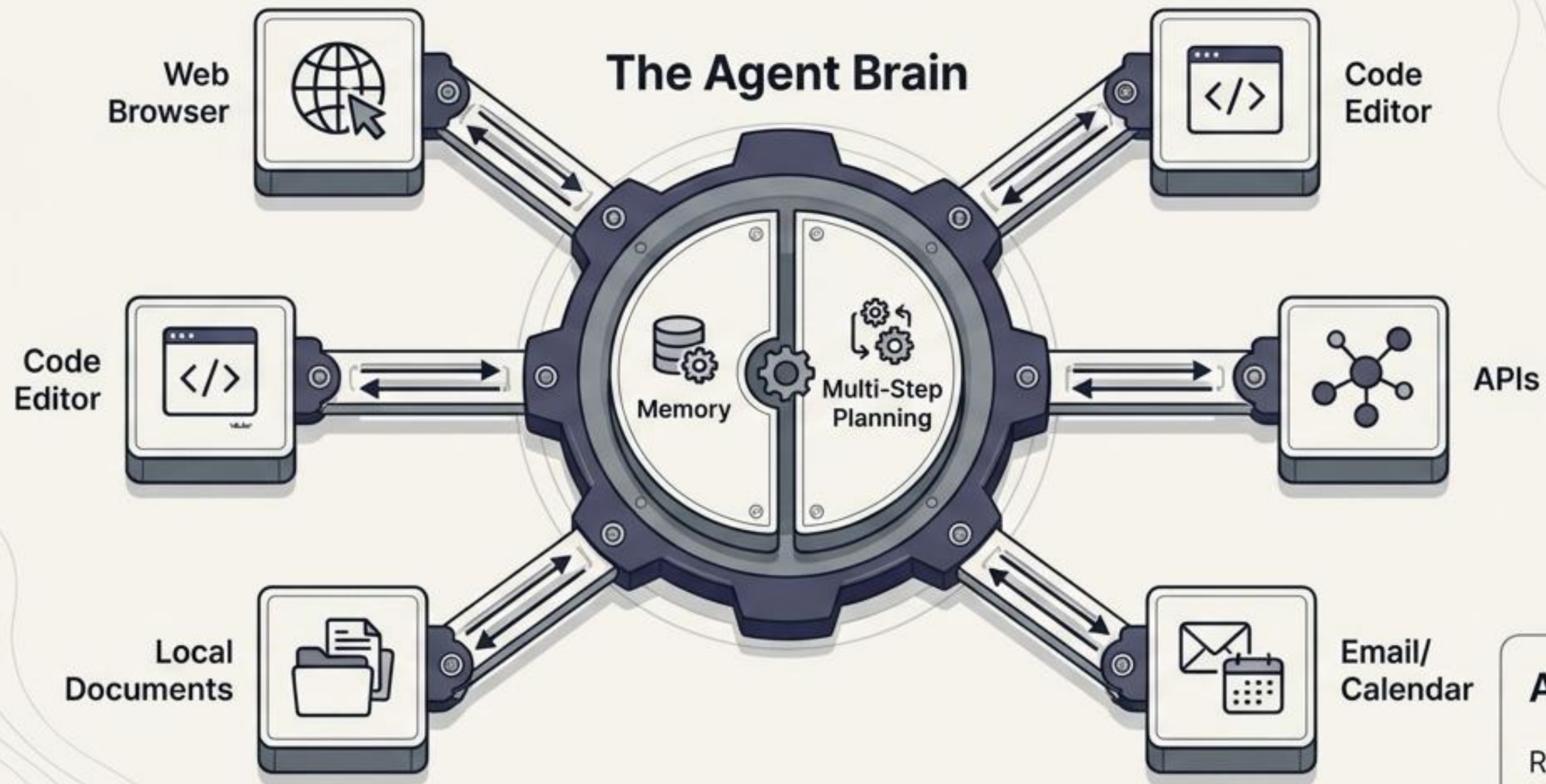
Capabilities Matrix

Predictive AI	
	Input: Structured Data
	Output: Forecasts & Classifications
	Autonomy: None (Tool)

Generative AI	
	Input: Natural Language Prompts
	Output: Novel Text, Code, & Media
	Autonomy: Low (Assistant)

AI Agents	
	Input: High-level Goals
	Output: Completed Multi-Step Actions
	Autonomy: High (Co-worker)

Autonomous agents utilize memory, planning, and external tools to execute goals.

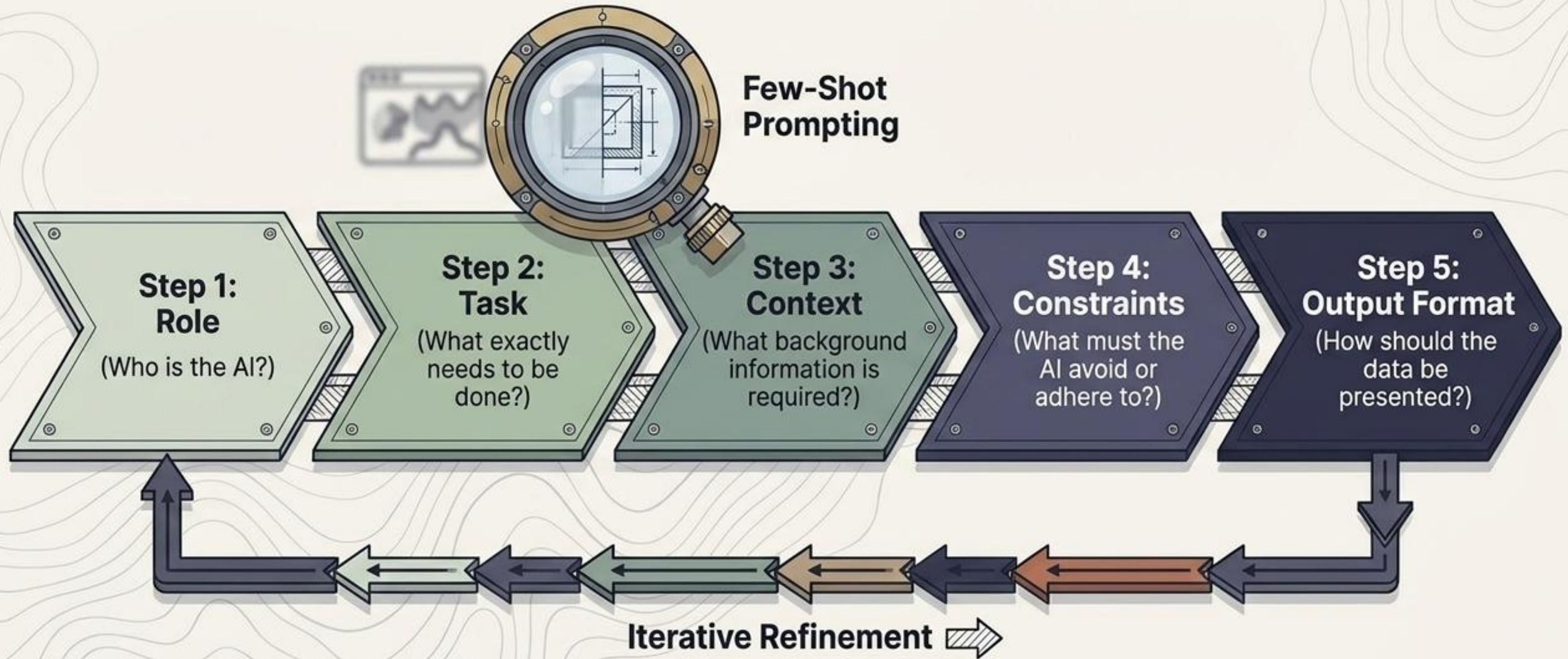


Agent Archetypes:

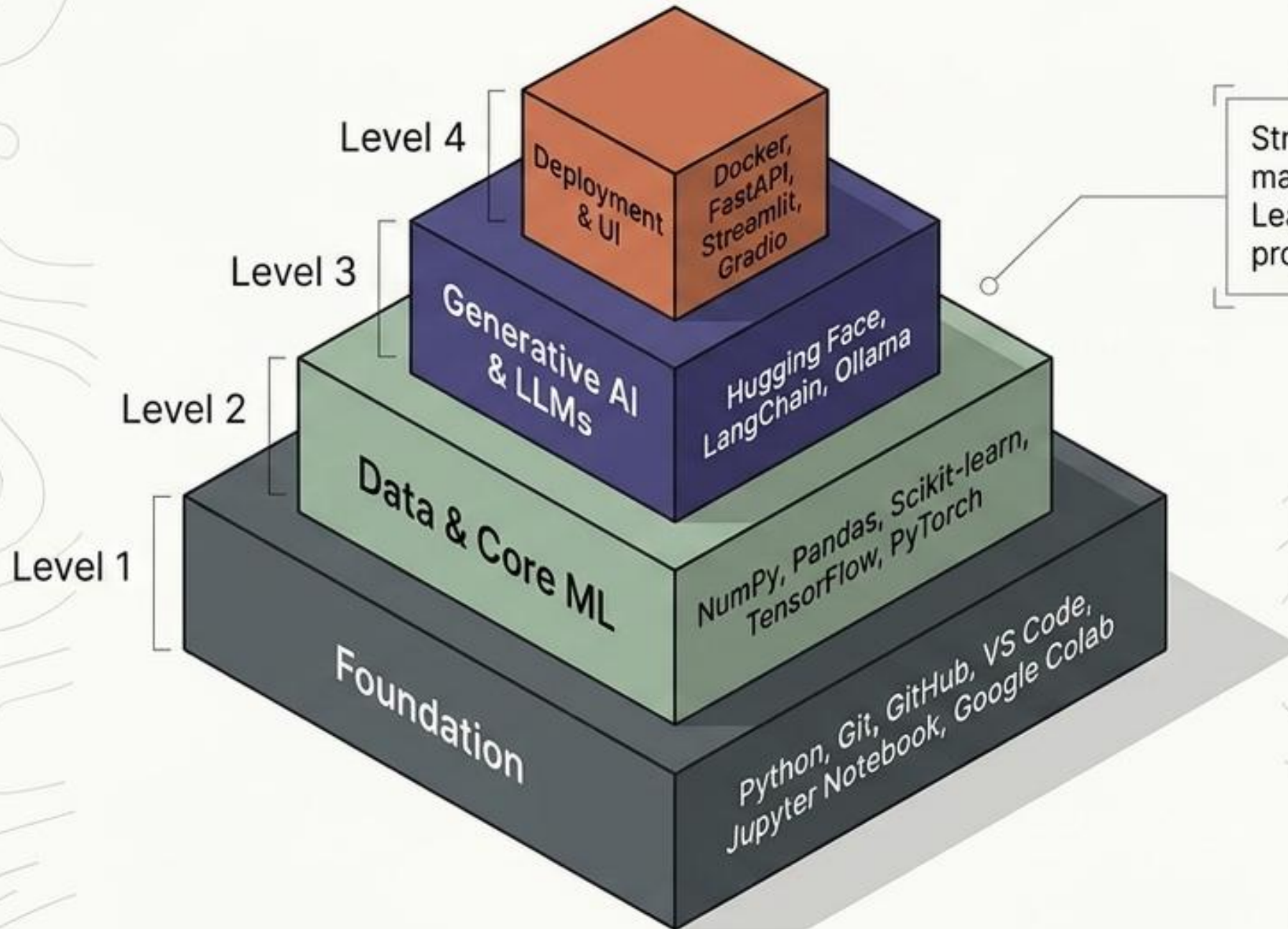
Research Agents,
Coding Agents,
Customer Support Agents,
Personal Productivity
Assistants

The Shift: From conversational chatbots to autonomous automation.

Effective prompt engineering operates as a structured communication pipeline.

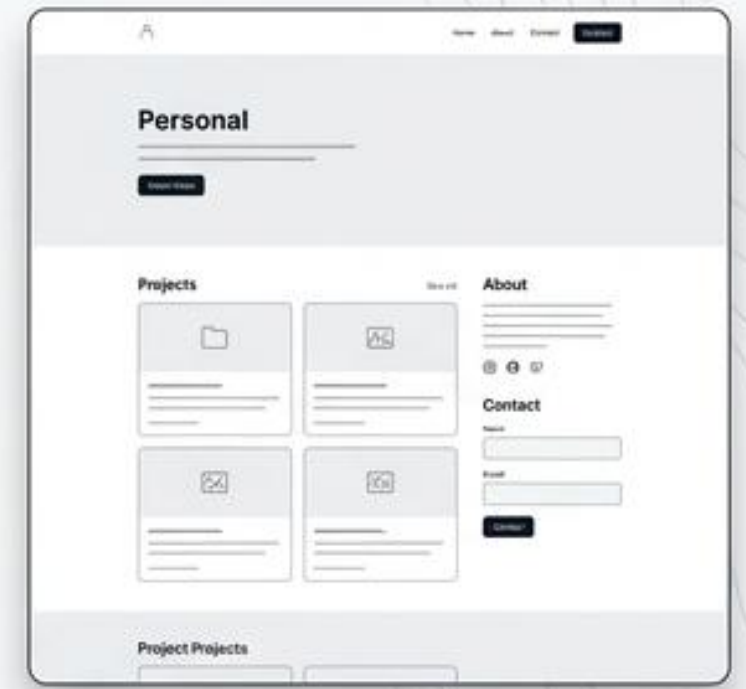
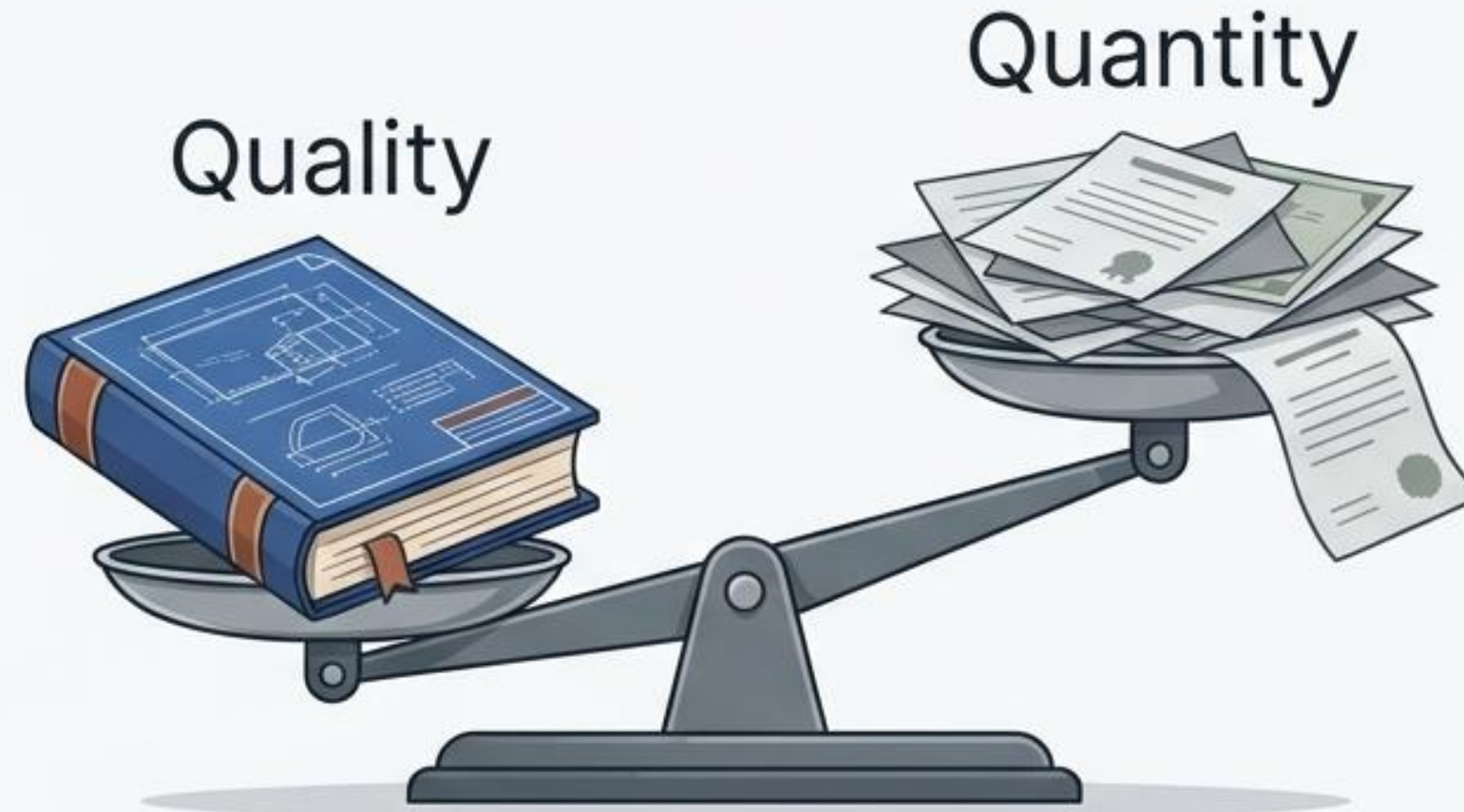
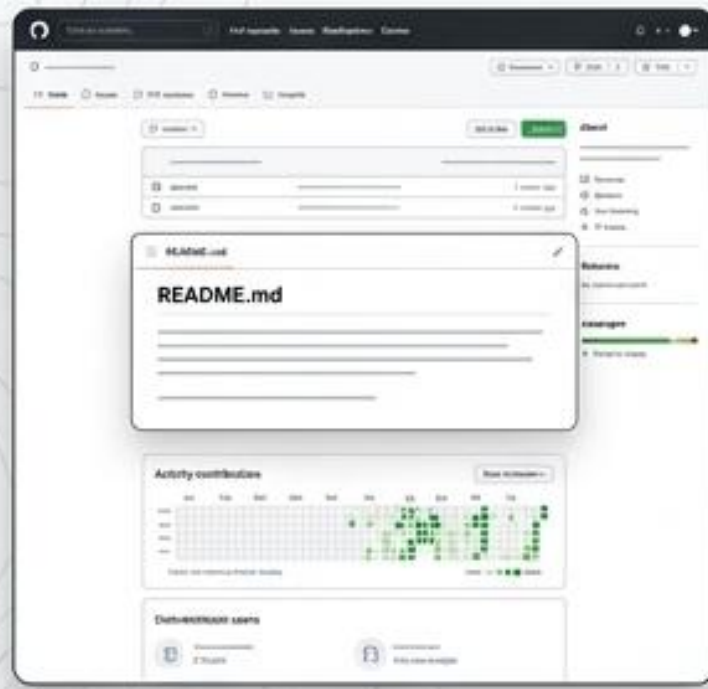


The modern AI technology stack is built in distinct, progressive layers.



Strategic Note: You don't need to master every tool immediately. Learn them gradually as your projects require them.

Recruiters prioritize deep, well-documented projects over a high quantity of certificates



Core Mandate: Showcase problem-solving. Write clear READMEs. Keep GitHub active.



AI Chatbot



Resume Analyzer



Image Classifier



Medical Diagnosis Assistant



Sentiment Analysis Dashboard



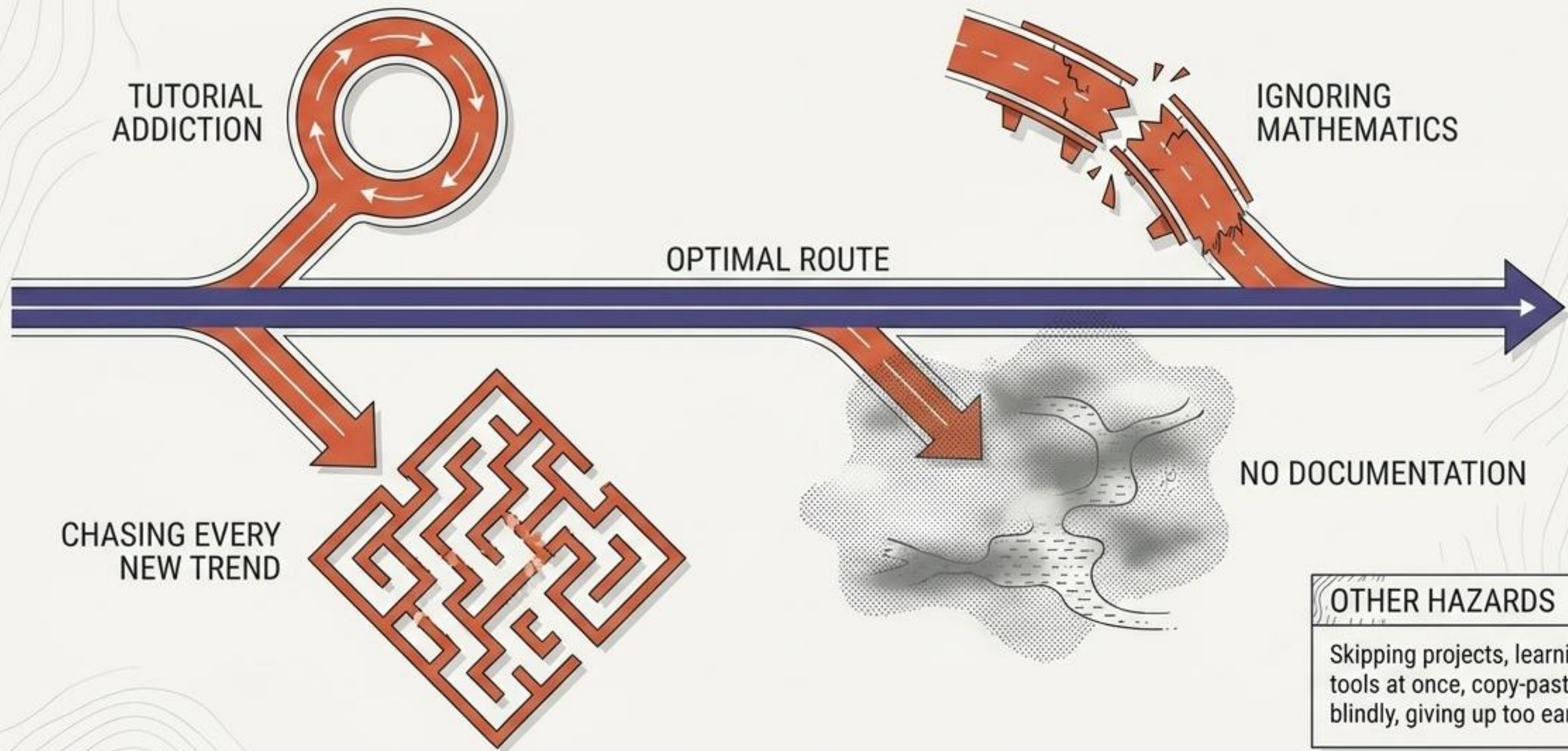
Recommendation System

Practical experience must bridge the gap between learning and employment.

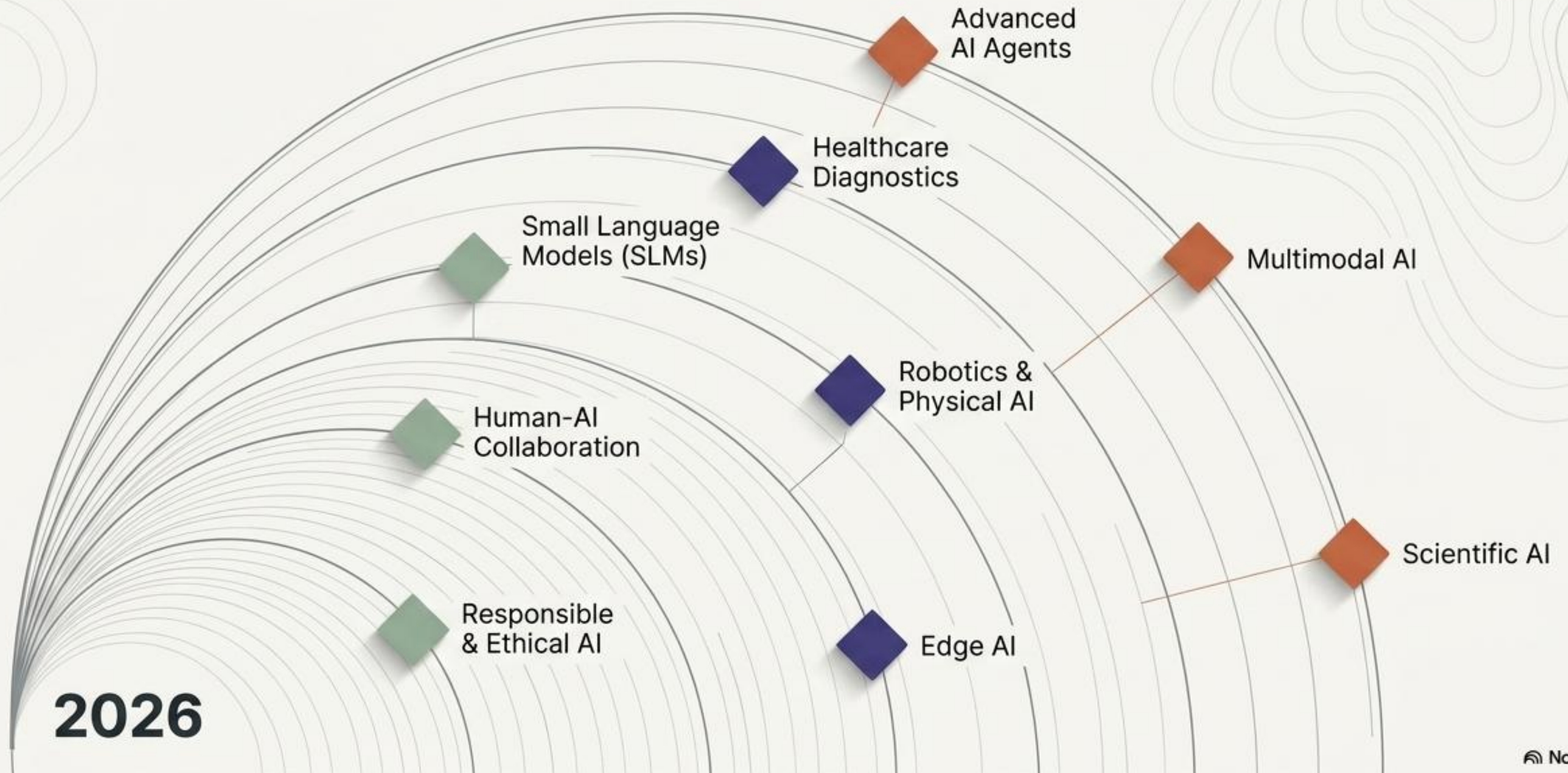


Key Insight: Courses build knowledge. Practical experience builds confidence.

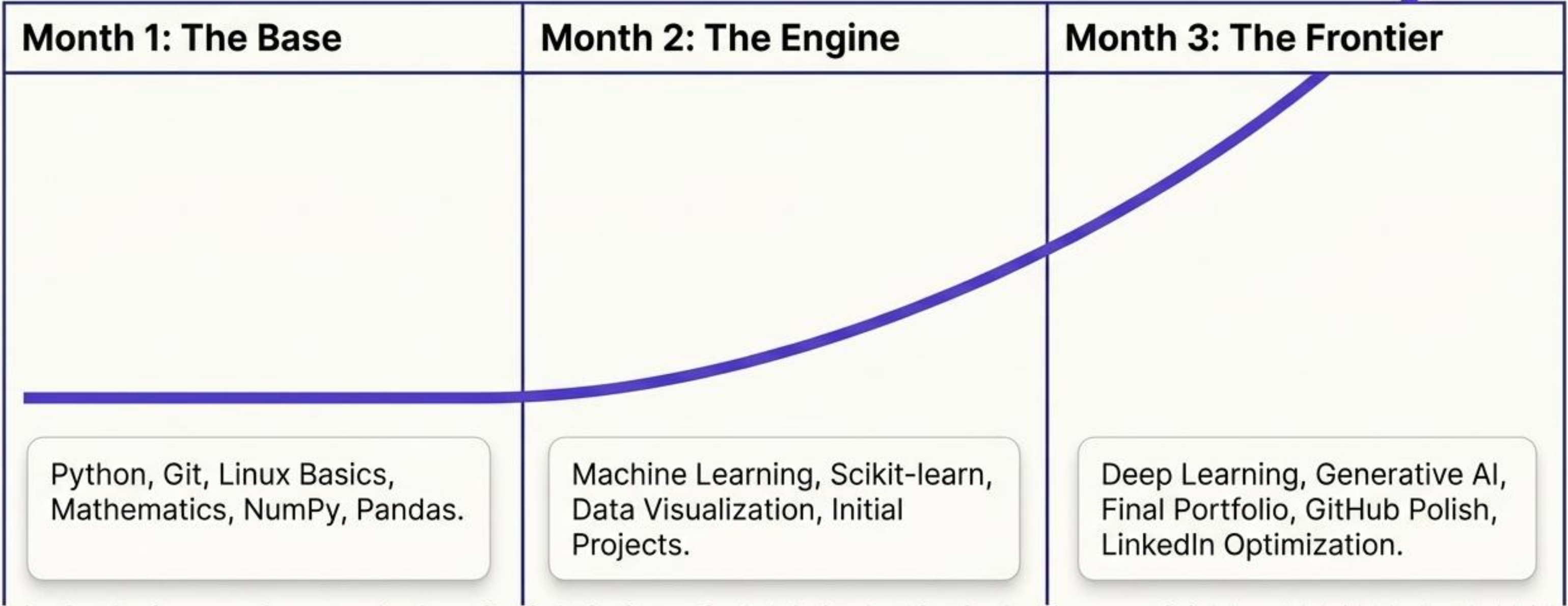
Navigating the AI landscape requires avoiding common beginner dead-ends



The future of artificial intelligence extends far beyond text generation.



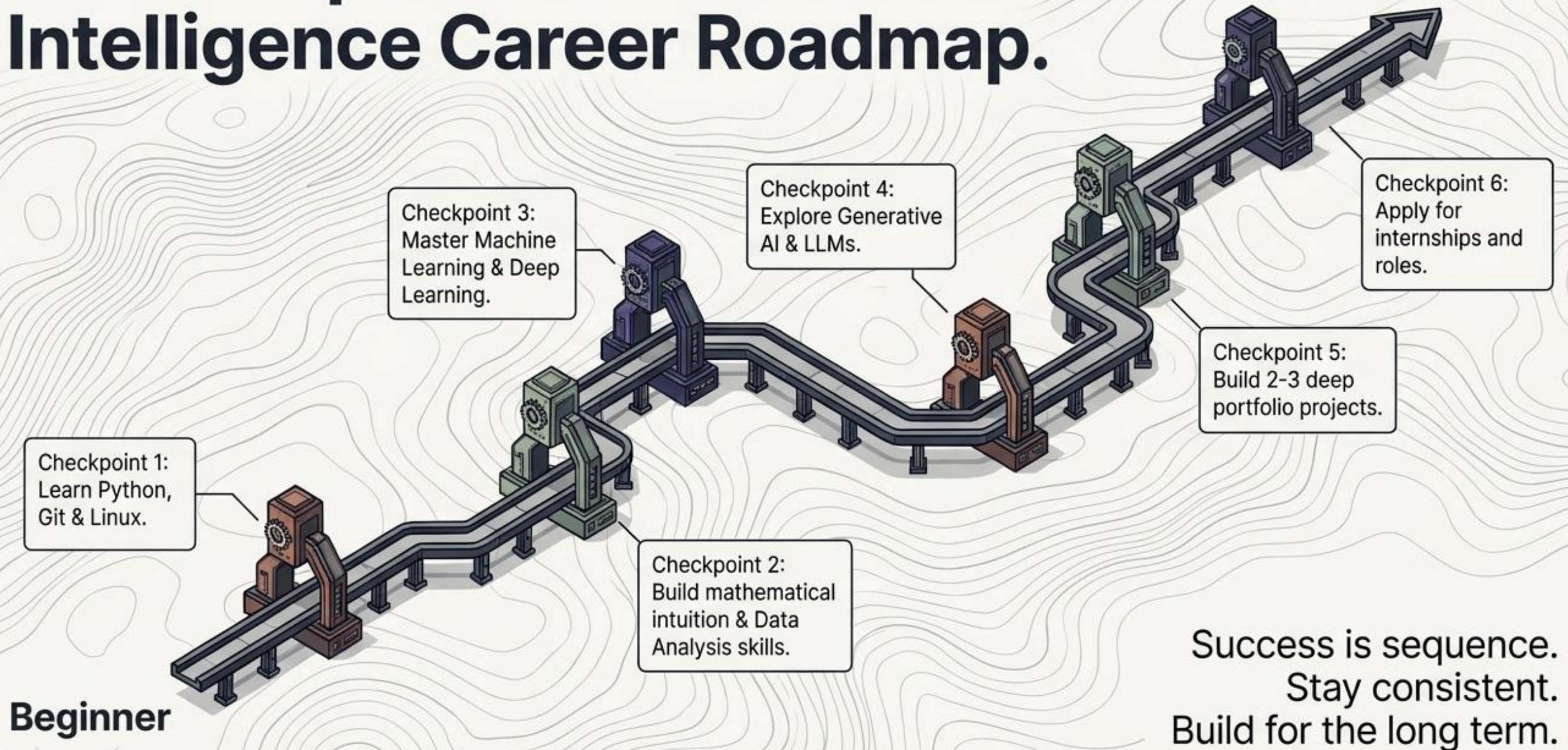
The 90-Day Execution Plan transforms theory into compounding daily momentum.



Execution Rule: Consistent daily practice > rushing through dozens of courses.

The Complete 2026 Artificial Intelligence Career Roadmap.

AI Professional





THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



PRESENTED BY

DAKSH KHADELWAL

FOUNDER, THINKHUB



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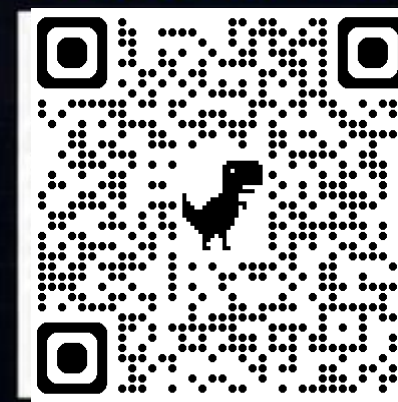


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KEEP LEARNING. KEEP BUILDING. KEEP GROWING.